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Agilex™ 7 FPGA I-Series Transceiver-SoC Development Kit User Guide

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1. Overview

Agilex™ 7 FPGA I-Series Transceiver-SoC Development Kit is a complete design environment that includes both hardware and software you need to develop Agilex 7 FPGA I-Series designs. The board provides a wide range of peripherals and memory interfaces to facilitate the development of Agilex 7 I-Series Transceiver-SoC designs.

Table 1. Ordering Information

Development Kit Version	Ordering Code	Device Part Number	Serial Number Identifier
Agilex 7 FPGA I-Series Transceiver-SoC Development Kit (Production 2 4× F-Tile) (Power Solution 2)	DK-SI-AGI027FD	AGIB027R31B1E1VC	AGI87TS4000001
Agilex 7 FPGA I-Series Transceiver-SoC Development Kit (Production 2 4× F-Tile) (Power Solution 2)	DK-SI-AGI027FC	AGIB027R31B1E1VB	3000001
Agilex 7 FPGA I-Series Transceiver-SoC Development Kit (Production 1 4× F-Tile) (Power Solution 2)	DK-SI-AGI027FA	AGIB027R31B1E1V	2000001
Agilex 7 FPGA I-Series Transceiver-SoC Development Kit (ES1 4× F-Tile) (Power Solution 1)	DK-SI-AGI027FB	AGIB027R31B1E1VAA	0001001
Agilex 7 FPGA I-Series Transceiver-SoC Development Kit (ES 4× F-Tile) (Power Solution 1)	DK-SI-AGI027FES	AGIB027R31B1E2VR0	0000001

For the board and FPGA capabilities, refer to the *Agilex 7 FPGA and SoC FPGA I-Series* page on the Altera website.

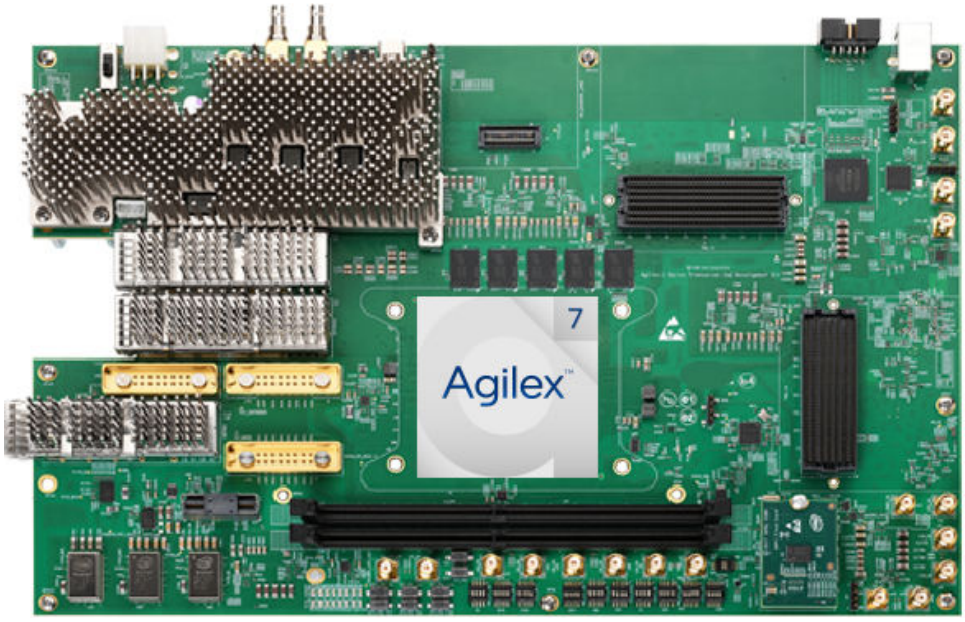
For more information about the *Agilex 7 Device Errata Sheet and User Guidelines (ES-1069)*, contact Altera Premier Support and quote ID #15011992053.

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*Other names and brands may be claimed as the property of others.

Figure 1. Agilex 7 FPGA I-Series Transceiver-SoC Development Kit (Power Solution 2 Board)—Top View



Figure 2. Agilex 7 FPGA I-Series Transceiver-SoC Development Kit (Power Solution 1 Board)—Top View



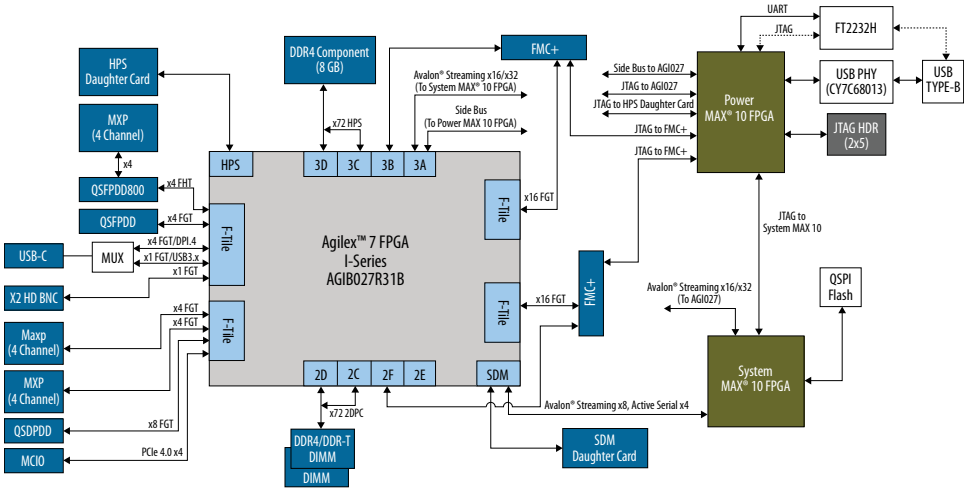
Refer to the *Appendix A—Development Kit Components* section for more details about the components on the Agilex 7 FPGA I-Series Transceiver-SoC Development Kit.

Related Information

- Development Kit Components on page 49
- Agilex 7 FPGA I-Series Transceiver-SoC Development Kit (4x F-Tile) Webpage
- Agilex 7 FPGA and SoC FPGA I-Series
- Agilex 7 F-Series and I-Series Known Issue List

1.1. Block Diagram

Figure 3. Agilex 7 FPGA I-Series Transceiver-SoC Development Kit Block Diagram



1.2. Feature Summary

- Agilex 7 FPGA I-Series, 2.7M LE, 3184B package
- F-Tile 1 (13C)
  - ☐ 4 FHT transceiver channels fan out to Quad Small Form Factor Double Density 800 (QSFPDD800)
  - ☐ 8 FGT transceiver channels to Quad Small Form Factor Double Density connector
  - ☐ 2 FGT transceiver channels to HD-BNC for SDI
  - ☐ 4 FGT transceiver as DP2.0(1)
  - ☐ 1FGT transceiver for USB3.2(1)
- F-Tile 2 (13A)
  - ☐ 4 FHT transceiver channels (116G) to MXPM
  - ☐ 8 FGT transceiver channels (56G) to MXPM
  - ☐ 8 FGT transceiver channels to Quad Small Form Factor Double Density connector
  - ☐ 4 FGT transceiver channels to MCIO (PCIe*)

- F-Tile 3 (12C)
  - ☐ 16 FGT channels to FMC+
- F-Tile 4 (12A)
  - ☐ 16 FGT channels to FMC+
- SR DDR4 2666 (x72 w/ ECC), 16GB
- 8GB SR DDR4-2666 (x72 w/ ECC) component down (HPS)
- IO48 interface for HPS Out of Box Experience (OOBE) daughter cards

1.3. Box Contents

- Agilex 7 FPGA I-Series Transceiver-SoC Development Kit
- Single-rank DDR4 DIMM module
- QSPI flash daughter card
- HPS IO48 OOBE daughter card
- NAND daughter card
- USB 2.0 Type B cable
- Ethernet cable
- 240W power adapter and NA/EU/JP/UK cords
- ATX power convert cable—24 pin to 6 pin

Related Information

Agilex 7 FPGA I-Series Transceiver-SoC Development Kit (4x F-Tile) Webpage

1.4. Recommended Operating Conditions

Table 2. Recommended Operating Conditions

Operating Condition	Range of Values
Ambient operating temperature range	0°C to 30°C
ICC load current	195 A
ICC load transient percentage	200 A/µs
FPGA maximum power supported by active heatsink/fan	160 W[2]

Related Information

Handling the Board on page 8

(1) Preliminary

(2) 160 W for core and 60 W for FGT transceiver.

2. Getting Started

2.1. Before You Begin

You must check the kit contents and inspect the boards to verify that you received all of the items in the box before using the kit and installing the software.

In case any of the items are missing, you must contact Altera before you proceed.

Important: Read the Appendix C.1—Safety and Regulatory Information for safe operation and regulatory adherence.

2.2. Handling the Board

When handling the board, it is important to observe static discharge precautions.

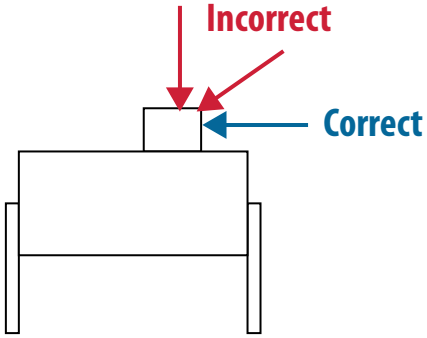
Caution: Without proper anti-static handling, the board can be damaged. Therefore, use anti-static handling precautions when touching the board.

Caution: This development kit should not be operated in a vibration environment.

2.2.1. Handling the DIP Switches

Operating the DIP Switch Safely

- Do not apply excessive force when operating the slide-type switch, as this might cause damage or deformation, leading to malfunction.
- Apply the operating load from the side of the striker.
- Avoid applying force at an angle or from above the striker, as this might deform the switch contact.



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Setting the DIP Switch

- To set the slide-type switch, use a small, rounded object like the end of a ballpoint pen or a tiny screwdriver.
- Do not use sharp tools, such as tweezers, to set the switch as they might damage it.
- Do not use a mechanical pencil, as its point could leave lead particles behind. These particles might interfere with the switch and the internal circuit board, potentially causing malfunction. Lead buildup can also reduce the dielectric strength of the circuit board.

2.3. Software and Driver Installation

This section explains how to install the following software and driver:

- Quartus® Prime Pro Edition software
- Agilex 7 FPGA I-Series Transceiver-SoC Development Kit software
- Altera® SoC FPGA Embedded Development Software (EDS)

The development kit includes onboard Altera FPGA Download Cable circuitry for programming the FPGA and the system MAX® 10 device. The Quartus Prime Pro Edition software installs the Altera FPGA Download Cable driver by default, so no separate driver installation is required.

Related Information

- Altera FPGA Software Installation and Licensing
- Altera FPGA Download Cable User Guide

2.3.1. Installing the Quartus Prime Pro Edition Software

1. Download the Quartus Prime Pro Edition software from the Quartus Prime Design Software web page of the Altera website.
2. Follow the on-screen instructions to complete the installation process. Choose an installation directory that is relative to the Quartus Prime Pro Edition software installation directory.

If you have difficulty installing the Quartus Prime software, refer to the *Altera FPGA Software Installation and Licensing*.

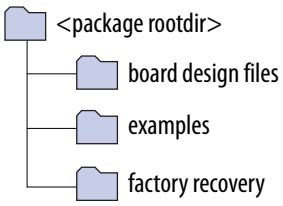
Related Information

- Quartus Prime Pro Edition User Guide: Getting Started
- Altera FPGA Software Installation and Licensing

2.3.2. Installing the Development Kit

1. Download the Agilex 7 FPGA I-Series Transceiver-SoC Development Kit installer package from the Agilex 7 FPGA I-Series Transceiver-SoC Development Kit (4x F-Tile) webpage on the Altera website.
2. Unzip the Agilex 7 FPGA I-Series Transceiver-SoC Development Kit installer package. The package creates the directory structure shown in the figure below.

Figure 4. Agilex 7 FPGA I-Series Transceiver-SoC Development Kit Directory Structure



3. For the latest issues and release notes, Altera recommends that you review the `readme.txt` located in the root directory of the kit installation.

Table 3. Installed Development Kit Directory Description

Directory Name	Description of Directory Contents
`board_design_files`	Contains schematic, layout, assembly, and Bill of Material (BOM) board design files. Use these files as a starting point for a new prototype board design.
`examples`	Contains the sample design files for the Agilex 7 FPGA I-Series Transceiver-SoC Development Kit: • Board Test System (BTS): BTS GUI, Clock GUI, and Power GUI • Golden Top project for pinout assignments management • Design Examples: Memory, XCVR, and GPIO
`factory_recovery`	Contains the original image/binary programmed onto the board before shipment. Use this image to restore the board with its original factory content.

2.3.3. Installing the Altera SoC FPGA Embedded Development Suite (EDS)

The Altera SoC FPGA Embedded Development Suite (SoC EDS) is a comprehensive software tool suite for embedded software development on Altera system-on-chip (SoC) devices. It contains development tools, utility programs, run-time software, and application examples to expedite firmware and application software of SoC embedded systems.

For more information and steps to install the Altera SoC EDS Tool Suite, refer to the links below.

Related Information

- Altera SoC FPGA Embedded Development Suite (SoC EDS) User Guide
- Arm* Development Studio for Altera SoC FPGA

3. Development Kit Setup

The instructions in this chapter explain how to setup the Agilex 7 FPGA I-Series Transceiver-SoC Development Kit for specific use cases.

3.1. Default Settings

The Agilex 7 FPGA I-Series Transceiver-SoC Development Kit ships with its board switches preconfigured to support the design examples in the kit. If you suspect your board might not be correctly configured with the default settings, follow the instructions in the factory default switch settings table to return to its factory settings before proceeding ahead.

Note: "X" refers to Don't Care in the table below.

Note: Do not set the switches when the power is on. Only set the switches after the power is off.

Table 4. Factory Default Switch Settings

Switch	Default Position	Default Function
S19[1:4]	OFF/OFF/ON/ON	System MAX® 10 and FPGA selected in the JTAG chain.
S20[1:4]	ON/ON/ON/ON	Mode 1: Onboard Altera download circuit act as the only JTAG master. Chained HPS with SDM nodes internally.
S9[1:4]	ON/OFF/OFF/X	Configuration mode setting bits: AS - Fast mode
S10[1:4]	ON/ON/ON/ON	SYS_SW[0:3] • SYS_SW[0]—Factory Loadn • '0'—Load image from Page 0 of the QSPI • SYS_SW[1]—MUX_SEL1 • SYS_SW[2]—MUX_SEL7 • SYS_SW[3]—MUX_SEL9

continued...

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Switch	Default Position	Default Function
S15[1:4]	ON/ON/ON/OFF	SYS_SW[4:7] • SYS_SW[4]—MUX_SEL_ZL • SYS_SW[5]—FMC-A PCIe RP/EP Select ☐ "0": RP (Default) ☐ "1": EP • SYS_SW[6]—FMC-B PCIe RP/EP Select ☐ "0": RP (Default) ☐ "1": EP • SYS_SW[7]—MCIO PCIe RP/EP Select ☐ "0": RP ☐ "1": EP (Default)
S1[1:4]	OFF/OFF/OFF/OFF	User Switch [0:3]
S6[1:4]	OFF/OFF/OFF/OFF	User Switch [4:7]
S22[1:4]	ON/ON/ON/ON	MUX_DIP_SW[0:3] • MUX_DIP_SW0—MUX_SEL2 • MUX_DIP_SW1—MUX_SEL3 • MUX_DIP_SW2—MUX_SEL4 • MUX_DIP_SW3—MUX_SEL5 Set to "ON" by default to select onboard clock as input.
S23[1:4]	ON/ON/ON/ON	MUX_DIP_SW[4:7] • MUX_DIP_SW4—MUX_SEL10 • MUX_DIP_SW5—MUX_SEL11 • MUX_DIP_SW6—MUX_SEL12 • MUX_DIP_SW7—MUX_SEL14 Set "0"/closed by default for onboard clock as input.
S4[1:4]	ON/ON/ON/ON	MUX_DIP_SW[8:11] • MUX_DIP_SW8—MUX_SEL13 • MUX_DIP_SW9—MUX_SEL6 • MUX_DIP_SW10—MUX_SEL0 • MUX_DIP_SW11—MUX_SEL8 Set "0"/closed by default for onboard clock as input.

3.2. Powering Up the Development Kit

1. Use the provided 240 W power adapter to supply power through J30.
2. After power adapter is plugged into J30 and switch S18 is set to the **ON** position, the LED D22 illuminates, indicating that the board power up successfully. If the LED D22 is not turned **ON**, it indicates that one or more power supply is incorrect.

3.3. Performing Board Restore

The Agilex 7 FPGA I-Series Transceiver-SoC Development Kit ships with bts_config design examples stored in the QSPI flash device. You can perform board restore by following the instructions through the Quartus Prime Programmer GUI.

3.3.1. Restoring Board System MAX 10 with Default Factory Image

1. Open the Quartus Prime Programmer GUI, and detect the JTAG chain after the system MAX 10 is restored.
2. Attach the system MAX 10 image on the system MAX 10 part.
3. Select programming options and click program button.

   Note: Once you plug the Altera FPGA Download Cable between `J11` and PC, the Altera onboard download cable circuit is disabled automatically.

3.3.2. Restoring Board QSPI Flash with Default Factory Image

Follow the steps to overwrite the Avalon® streaming interface x8 configuration mode image.

1. Ensure that MSEL[2:0] is OFF (JTAG mode) before powering up the board.
2. Open Quartus Prime Programmer GUI, detect JTAG chain.
3. Change VTAP10 to 10M50DAF256, attach flash device of QSPI_2Gb.
4. Attach the Avalon streaming interface x8 configuration mode image to QSPI_2Gb.
5. Select program/configure options and click **Start** button.
6. Power off the board after programming the flash successfully, set MSEL[2:0] OFF OFF ON (Avalon streaming interface x8 configuration mode).

4. Board Test System

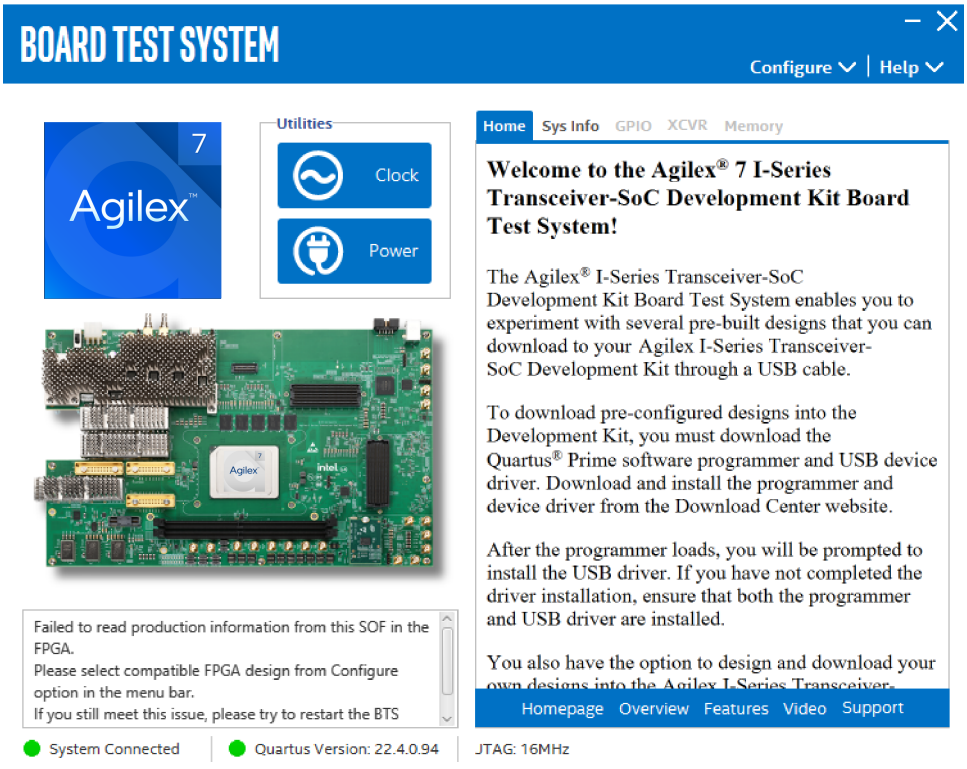
The Agilex 7 FPGA I-Series Transceiver-SoC Development Kit includes design examples and the board test system (BTS) GUI to test the functionality of this board. The BTS provides an easy-to-use interface to alter functional settings and observe results. You can use the BTS to test board components, modify functional parameters, observe performance, and measure power usage.

While using the BTS, you reconfigure the FPGA several times with test designs specific to the functionality that you are testing. The BTS is also useful as a reference for designing systems. The BTS communicates over the JTAG bus to a test design running in the Agilex 7 FPGA I-Series.

The following figure shows the graphical user interface (GUI) of a board that is in factory configuration.

Figure 5. BTS GUI (Power Solution 2 Board—Non-Enpirion*)

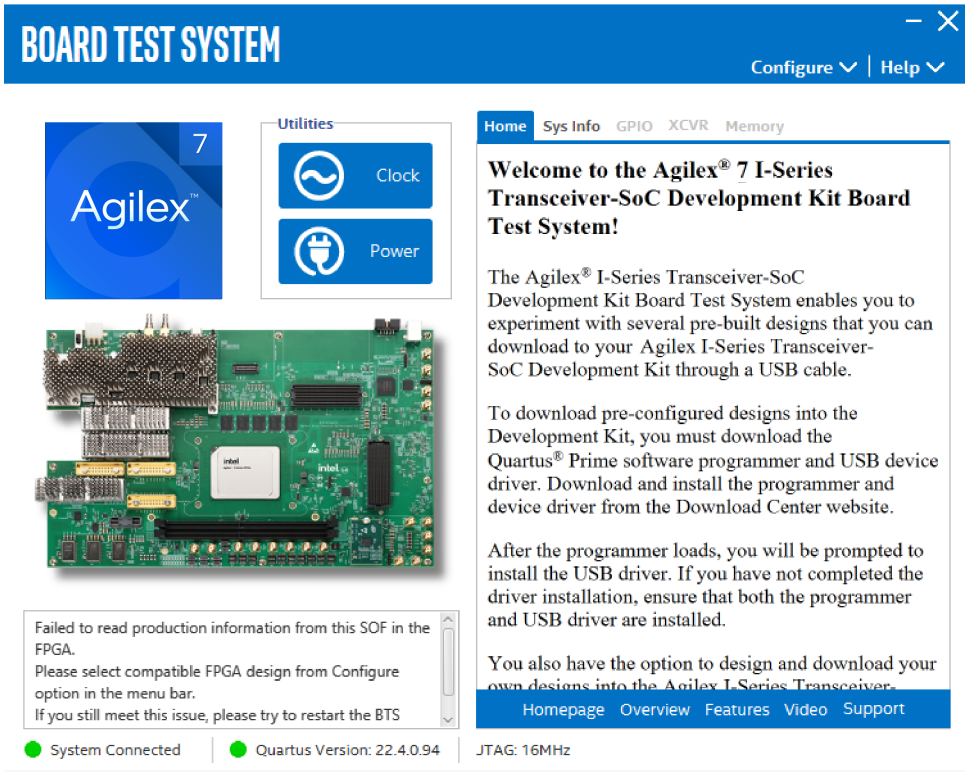
This figure applies to DK-SI-AGI027FD, DK-SI-AGI027FC, and DK-SI-AGI027FA.



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Figure 6. BTS GUI (Power Solution 1 Board—Enpirion*)

This figure applies to DK-SI-AGI027FB and DK-SI-AGI027FES.



4.1. Set Up BTS GUI Running Environment

You need to download and install **Java Runtime Environment** and Quartus Prime software on your systems to run the BTS GUI, including the Power Monitor and Clock Controller GUI. **Java Runtime Environment** includes **OpenJDK** and **OpenJFX**, whose installation is a one-time procedure. If you have completed it before, you do not need to do it again unless the Java version upgrade is needed.

Note: Installing the **Java Runtime Environment** requires Administrator privileges on Windows* and root privileges on Linux*.

4.1.1. Downloading OpenJDK

1. Download the Temurin* OpenJDK. Refer to the related information for the download link.
2. Select Architecture x64, Package Type JRE, and Version 11.
   a. For the **Windows** system, choose the `JRE.zip` format file.
   b. For the **Linux** system, choose the `JRE.tar.gz` format file.

   Note: Download the latest version to update the JDK version. The following version was tested: 11.0.21+9

Related Information

Temurin* OpenJDK

4.1.2. Downloading OpenJFX

1. Download the Gluon* OpenJFX. Refer to the related information for the download link.
2. Select JavaFX version 17.0.2.
   a. For the **Windows** system, download the JavaFX Windows x64 SDK.
   b. For the **Linux** system, download the JavaFX Linux x64 SDK.

Related Information

Gluon* OpenJFX

4.1.3. Installing OpenJDK and OpenJFX

You have two downloaded compressed files. Follow these steps to install them.

Follow these steps to install the downloaded zip files:

1. On a **Windows** system, ensure that you have administrator privileges. The Altera-provided BTS execution script (`BoardTestSystem.bat`) requires OpenJDK (JRE) and OpenJFX to be unzipped, installed, and renamed in the specified directories .Administrator access is required to perform these actions.
   a. Unzip the files and put them in the following directory:
      - `C:\Program Files\Java\jre`
      - `C:\Program Files\Java\jfx`
   b. The unzipped folder name of JRE is `jdk-11.0.xx+x-jre` (for example, `jdk-11.0.15+10-jre`). Rename it to *jre*.
   c. The unzipped folder name of JFX is `javafx-sdk-17.0.2`. Rename it to *jfx*.
2. On a **Linux** system, the Altera-provided BTS execution script (`BoardTestSystem.sh`) requires OpenJDK (JRE) and OpenJFX to be unzipped, installed, and renamed in the specified directories. Unzip the files and rename the folders using the following commands:

```
# unzip openjfx-17.0.2_linux-x64_bin-sdk.zip -d /opt/Java/
# tar zxvf OpenJDK11U-jre_x64_linux_hotspot_11.0.15_10.tar.gz -C /opt/Java/
# cd /opt/Java
# mv javafx-sdk-17.0.2 jfx
# mv jdk-11.0.15+10-jre jre
```

   You have the following two directories on your **Linux** system:
   - `/opt/Java/jre`
   - `/opt/Java/jfx`

3. On the **Linux** system, add current user to the group `dialout` to ensure the access to UART, using the following commands:

```
ls -l /dev/ttyUSB*

sudo usermod -a -G dialout $USER
groups
```

Note: You must reboot the system for the UART access settings to take effect.

4.1.4. Setting Up the Quartus Prime Software for BTS Operation

You must install the Quartus Prime software to support the silicon on the development kit. The recommended version is located in the `README.txt` file in the `<*package dir*>\examples\board_test_system` directory. If you choose to install individual files, you must install Agilex 7 Device Support.

The BTS communicates over JTAG to a test design running in the FPGA. The BTS shares the JTAG with other applications such as the Programmer, the System Console, and the Signal Tap Logic Analyzer. Altera recommends closing other applications before using BTS, as the GUI is designed based on the Quartus Prime software.

The BTS relies on the Quartus Prime software's specific library. Before running the BTS, open the Quartus Prime software to automatically set the environment variable `QUARTUS_ROOTDIR`. You can also change it through **Environment Variables** in the **System Properties** in Windows*. The BTS uses this environment variable to locate the Quartus Prime library.

4.1.5. Running BTS GUI

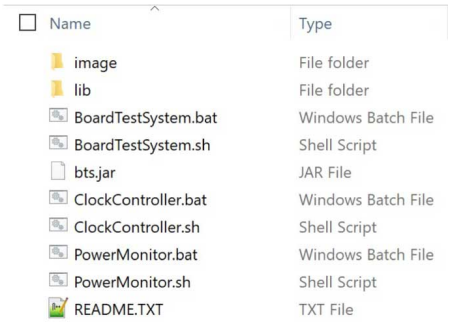
With the power to the board off, follow these steps:

1. Connect the USB cable to your PC and the board.
2. Check the development board switches and jumpers are set according to your preferences. Refer to the the Development Kit Setup section. In most cases, BTS requires the system MAX 10 and Agilex 7 FPGA on the JTAG chain while the Clock Controller and Power Monitor require only the system MAX 10.
3. Check the external modules status: MXPM cable/QSFP/QSFPDD/FMC/DIMM.
4. Turn on the board power switch.

Note: To ensure operating stability, keep the USB cable connected and the board powered on when running the demonstration application. You must attach the USB cable and power on the board for BTS to run correctly.

Navigate to the `<*packagedir*>\examples\board_test_system` directory to run BTS. The BTS release folder always includes the following files.

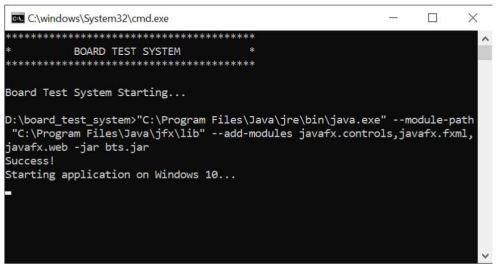
Figure 7. BTS Folder



You can run BTS GUI easily with the following scripts.

1. On a **Windows** system, select `BoardTestSystem.bat` and run as an administrator to run BTS, Clock Controller, or Power Monitor GUI.

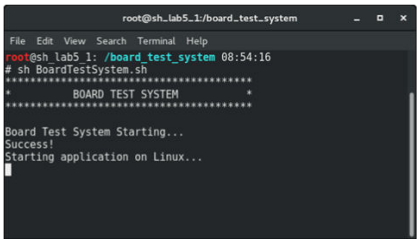
Figure 8. Windows Console



2. On a **Linux** system, make the shell script executable and run it with root privileges using the following command:

```
chmod +x ./BoardTestSystem.sh
```

Figure 9. Linux Console



Note: The `.bat` or shell script checks the Java environment settings, copies necessary files, and gives some prompts if the environment is not set up correctly.

The GUI displays the application tab corresponding to the design running in the FPGA. If the design loaded in the FPGA is not supported, it prompts a message to configure your board with a valid BTS design. Refer to the The Configure Menu section.

4.2. Test the Functionality of the Development Kit

This section describes each control in the BTS.

4.2.1. The Bottom Information Bar

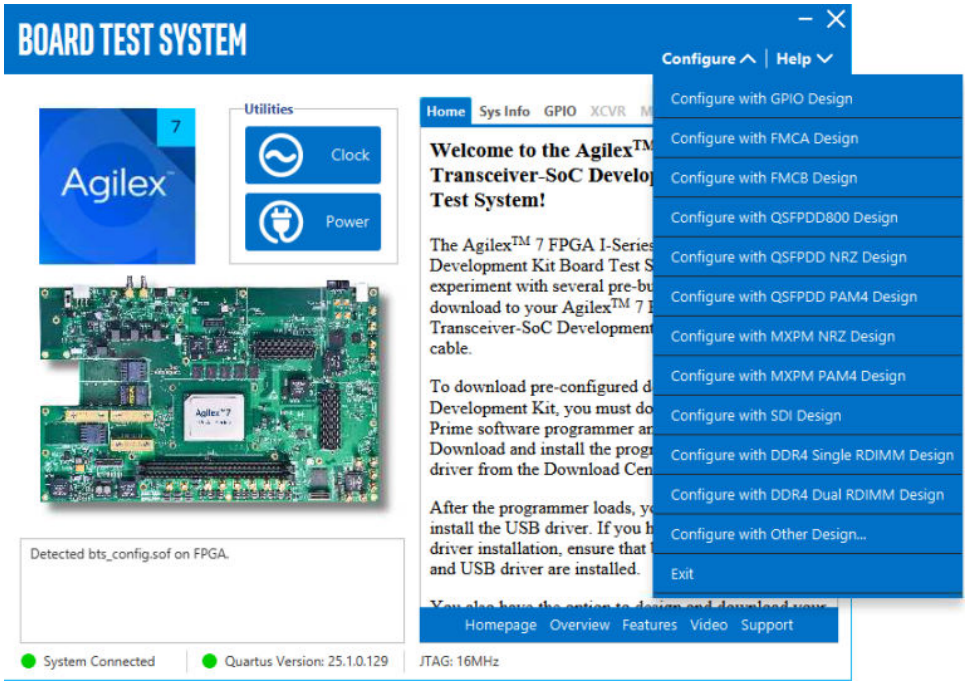
The bottom information bar shows the status of the system connection, the Quartus Prime version and the JTAG clock.

- **System Connected/Disconnected**: Shows if the board is connected to the system. The green sign turns gray if the board becomes disconnected.
- **Quartus Prime Version**: Displays the current Quartus Prime version installed and active on your system. The text turns red if your version is older than the required version. Change the *QUARTUS_ROOTDIR* environment variable if you have installed the right version but the active version does not meet the requirement.
- **JTAG**: Displays the JTAG clock frequency.

4.2.2. The Configure Menu

Use the **Configure** menu to select the design you want to use. Each design example tests different functionality that corresponds to one or more application tabs.

Figure 10. The Configure Menu



To configure the FPGA with a test system design, follow these steps:

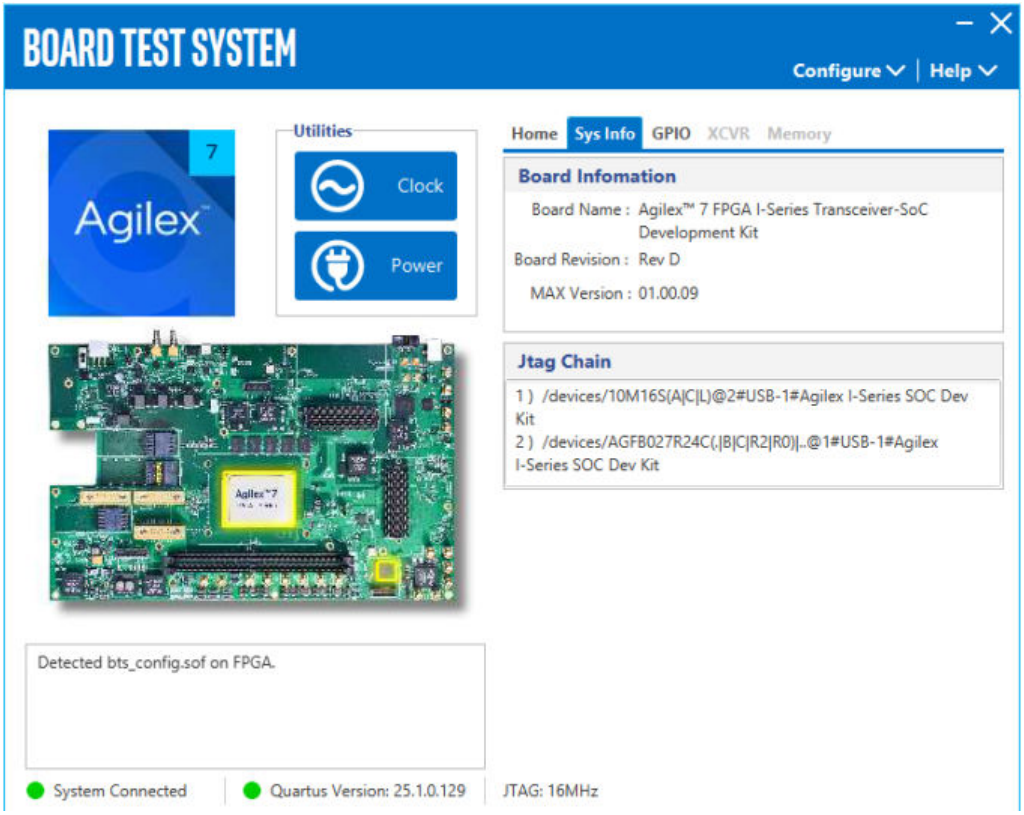
1. On the **Configure** menu, click the **Configure** command that corresponds to the functionality you want to test.
2. In the dialog box that appears, click **Configure** to download the corresponding design's SRAM Object File (`.sof`) to the FPGA. The download process usually takes less than a minute.

When configuration is completed, the design begins running in the FPGA. The corresponding GUI application tabs that interface with the design is now enabled. If you use the Quartus Prime Programmer for configuration, instead of the BTS GUI, you might need to restart the GUI.

4.2.3. The Sys Info Tab

The **Sys Info** tab shows information about the board's current configuration. The tab displays the board information, JTAG Chain devices and other details stored on the board.

Figure 11. The Sys Info Tab



The following sections describe the controls on the System Info tab.

Board Information

The board information control displays static information about your board.

- **Board Name**: Indicates the official name of the board given by the BTS.
- **Board Revision**: Indicates the revision of the board.
- **MAX Version**: Indicates the version of the system MAX 10.

JTAG Chain

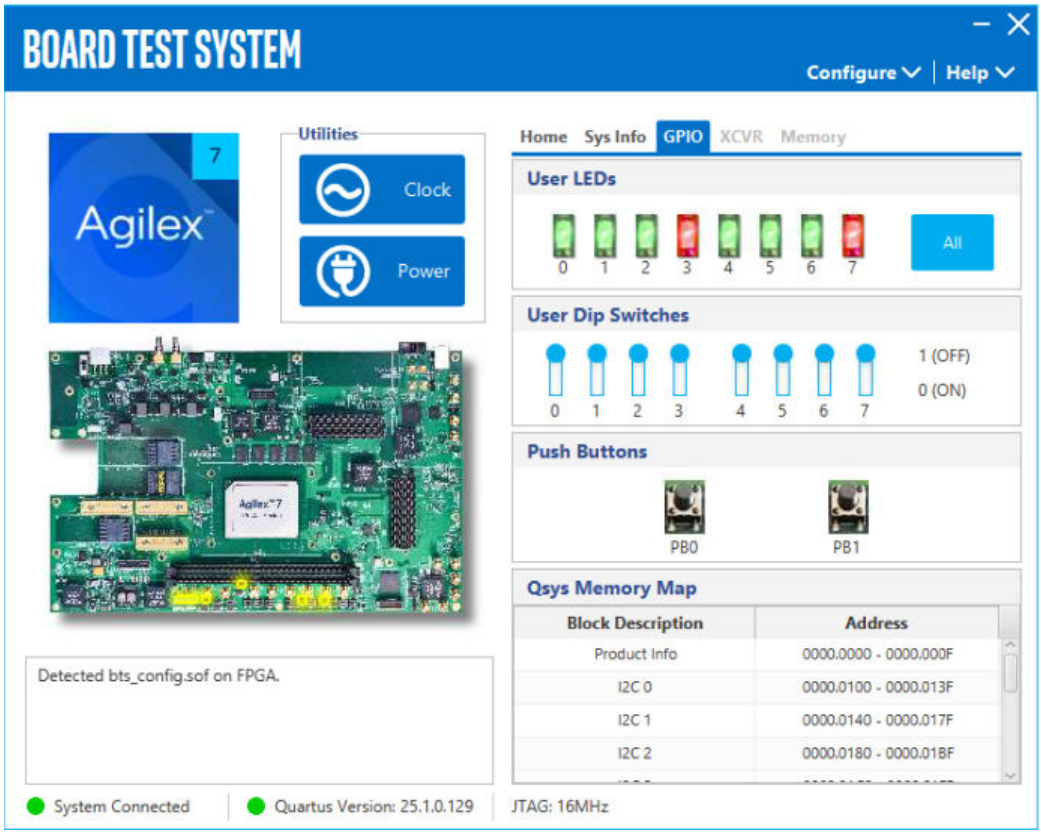
The JTAG chain control shows all the devices currently in the JTAG chain.

Note: You should place the system MAX 10 and FPGA in the JTAG chain when running the BTS GUI.

4.2.4. The GPIO Tab

The **GPIO** tab allows you to interact with all the general-purpose user I/O components on your board. You can turn LEDs on or off and see the status of push buttons and DIP switches.

Figure 12. The GPIO Tab



The following sections describe the controls on the **GPIO** tab.

User LEDs

The **User LEDs** control displays the current state of the user LEDs. Toggle the LED buttons to turn the board LEDs on and off. Click the All button to reverse the state of all the LEDs.

User Dip Switches

The **User Dip Switches** control display the status of the USER_SW[0:3] (`S1`) and USER_SW[4:7] (`S6`).

Push Buttons

The **Push Buttons** control shows the status of PB0 and PB1.

Qsys Memory Map

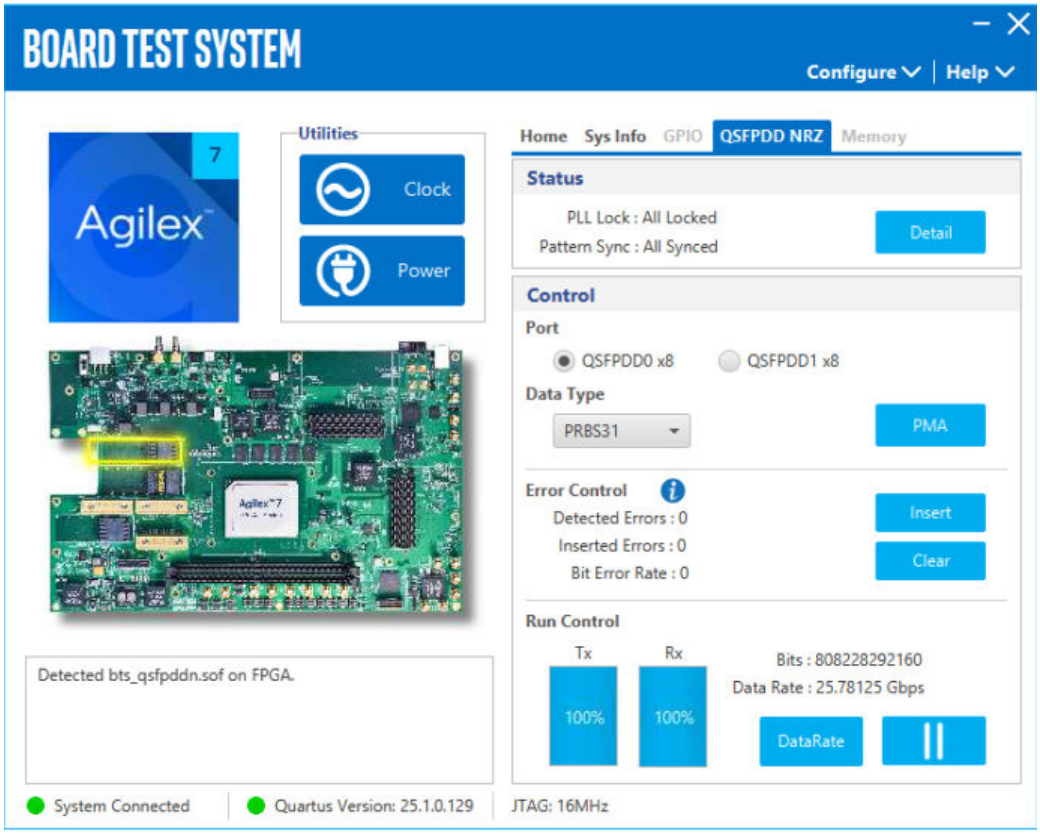
The **Qsys Memory Map** control shows the memory map of `bts_config.sof` design running on your board.

4.2.5. The XCVR Tab

The **XCVR** tab allows you to run transceiver tests on your board. You can run the test using either electrical loopback modules or optical fiber modules.

4.2.5.1. The QSFPDD NRZ Tab

Figure 13. The QSFPDD NRZ Tab



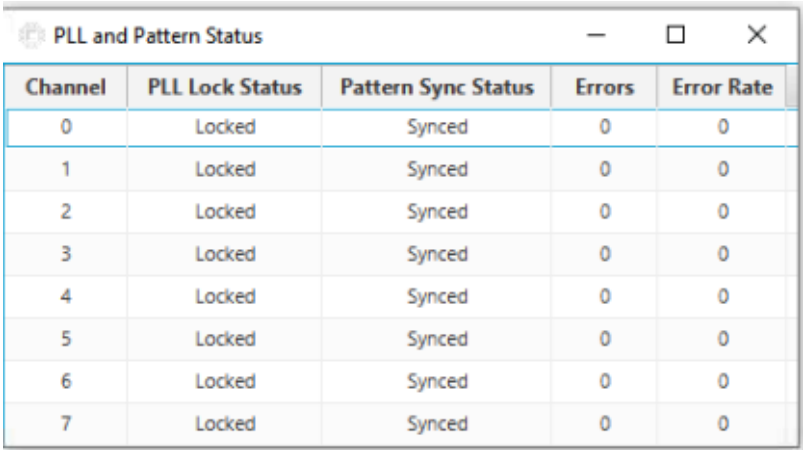
The following sections describe controls in the **QSFPDD NRZ** tab.

Status

The Status control displays the following status information during the loopback test:

- **PLL lock**: Shows the PLL locked or unlocked state.
- **Pattern Sync**: Shows the pattern synced or not state. The pattern is considered synced when the start of the data sequence is detected.
- **Detail**: Shows the PLL lock and pattern sync status of each channel. The number of the error bits of each channel can be found here.

Figure 14. QSFPDD NRZ—PLL and Pattern Status



Channel	PLL Lock Status	Pattern Sync Status	Errors	Error Rate
0	Locked	Synced	0	0
1	Locked	Synced	0	0
2	Locked	Synced	0	0
3	Locked	Synced	0	0
4	Locked	Synced	0	0
5	Locked	Synced	0	0
6	Locked	Synced	0	0
7	Locked	Synced	0	0

Control

Use the following controls to select an interface to apply PMA settings, data type, and error control:

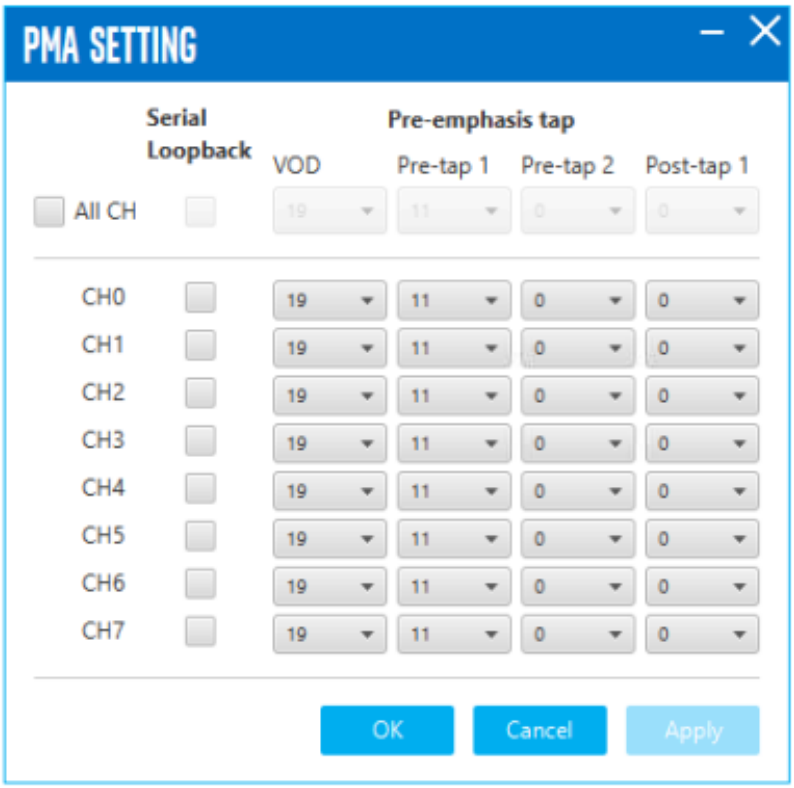
- **QSFPDD0 x8**
- **QSFPDD1 x8**

PMA Setting

Allows you to make changes to the PMA parameters that affect the active transceiver interface. The following settings are available for analysis:

- **Serial Loopback:** Displays the signal status between the transmitter and the receiver.
- **VOD**: Specifies the voltage output differential of the transmitter buffer.
- **Pre-emphasis tap**:
  - **Pre-tap 1**: Specifies the amount of pre-emphasis on the first pre-tap of the transmitter buffer.
  - **Pre-tap 2**: Specifies the amount of pre-emphasis on the second pre-tap of the transmitter buffer.
  - **Post-tap 1**: Specifies the amount of pre-emphasis on the post-tap of the transmitter buffer.

Figure 15. QSFPDD NRZ—PMA Setting



Data Type

The Data Type control specifies the type of data pattern contained in the transactions. Select the following available data types for analysis:

- **PRBS7**: Pseudo-random 7-bit binary sequences
- **PRBS15**: Pseudo-random 15-bit binary sequences
- **PRBS23**: Pseudo-random 23-bit binary sequences
- **PRBS31**: Pseudo-random 31-bit binary sequences (default)

Error Control

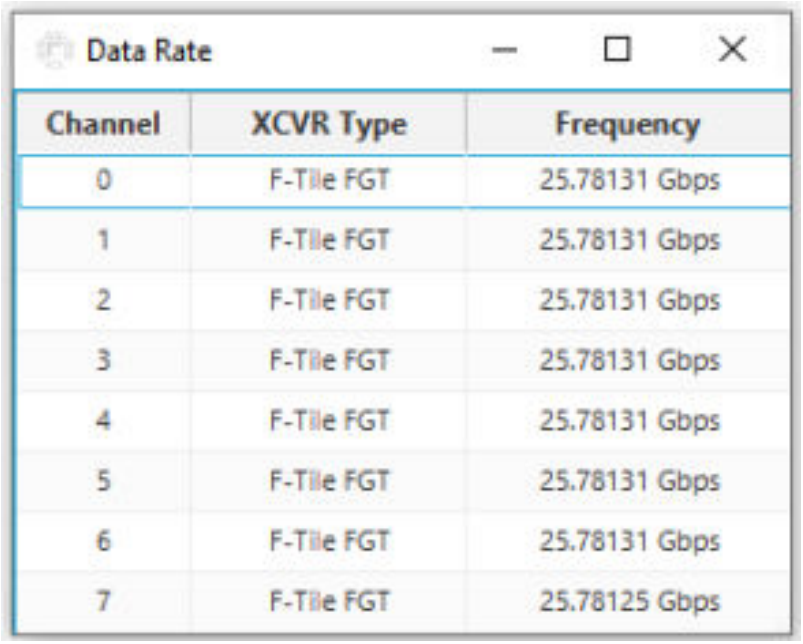
This control displays data errors detected during analysis and allows you to insert errors:

- **Detected Errors**: Displays the number of data errors detected in the received bit stream.
- **Inserted Errors**: Displays the number of errors inserted into the transmit data stream.
- **Bit Error Rate**: Calculates the bit error rate of the transmit data stream.
- **Insert**: Insert a one-word error into the transmit data stream each time you click the button. **Insert** is only enabled during transaction performance analysis.
- **Clear**: Resets the **Detected Errors** counter and **Inserted Errors** counter to zeros.

Run Control

- **TX** and **RX** performance bars: Show the percentage of maximum theoretical data rate that the requested transactions are able to achieve.
- **Start**: This control initiates the loopback tests.
- **Data Rate**: Displays the XCVR type and data rate of each channel.

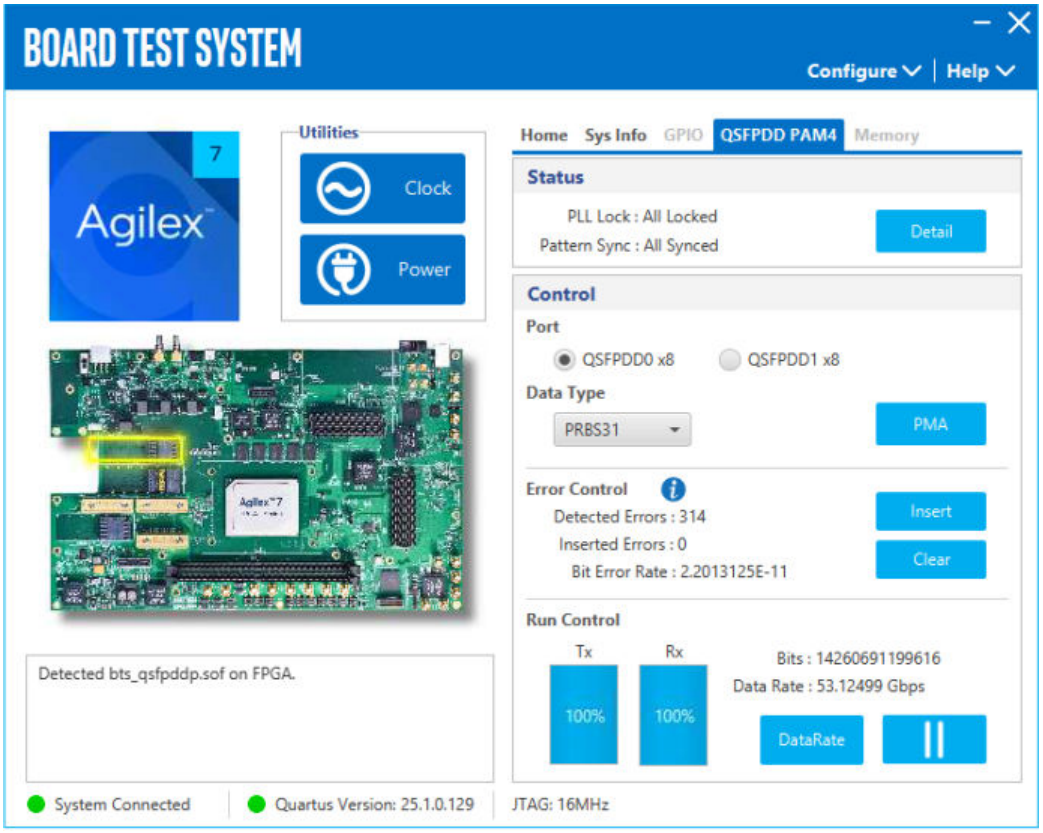
Figure 16. QSFPDD NRZ—Data Rate



4.2.5.2. The QSFPDD PAM4 Tab

This tab has similar control functions with the **QSFPDD NRZ** tab.

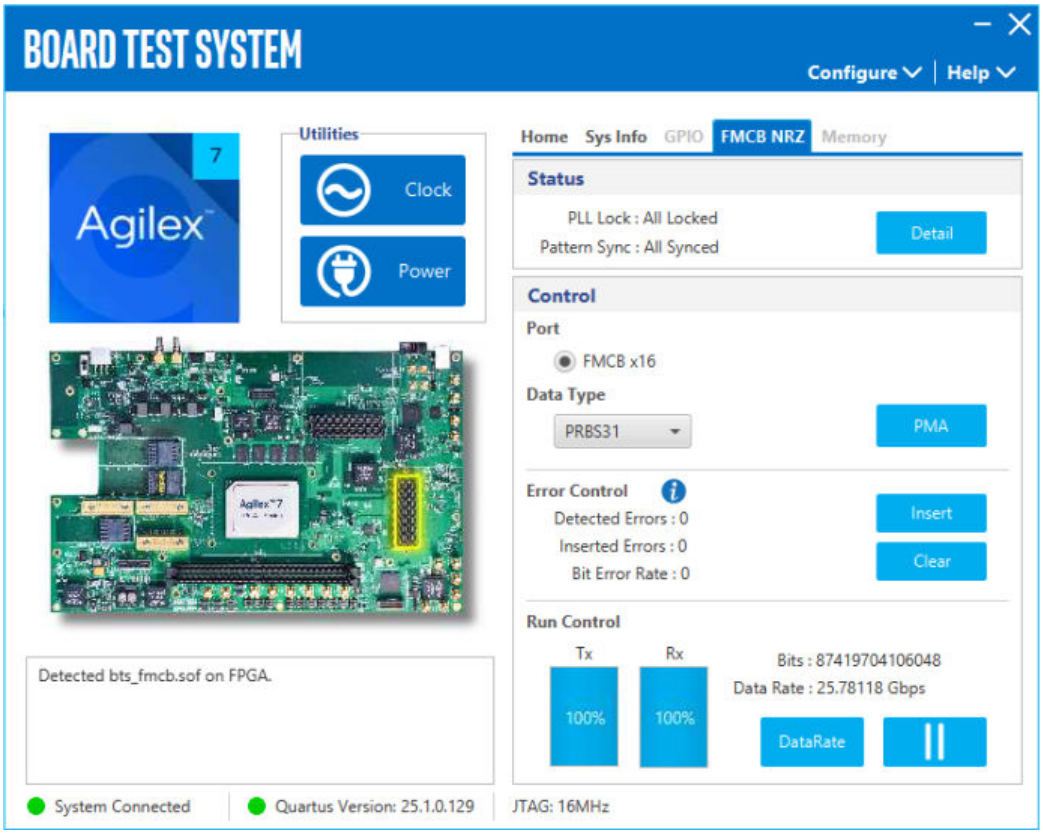
Figure 17. The QSFPDD PAM4 Tab



4.2.5.3. The FMCB NRZ Tab

This tab has similar control functions with the **QSFPDD NRZ** tab except for the port selection.

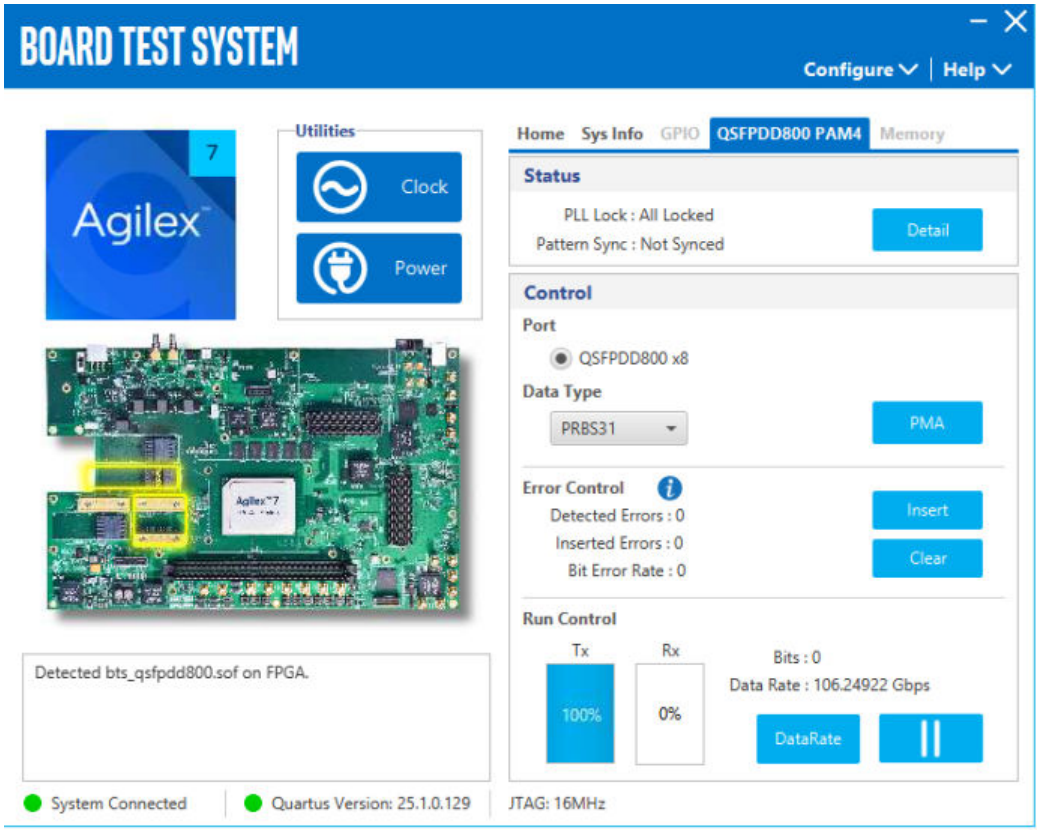
Figure 18. The FMCB NRZ Tab



4.2.5.4. The QSFPDD800 PAM4 Tab

Similar control functions with the **QSFPDD NRZ** tab.

Figure 19. The QSFPDD800 PAM4 Tab

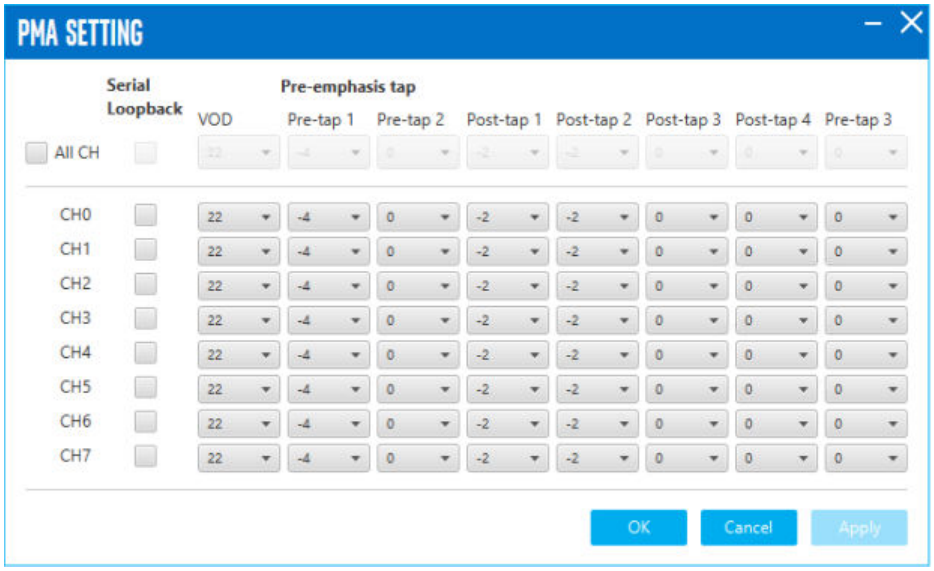


PMA Setting

In addition to the PMA settings listed in the *QSFPDD NRZ—PMA Setting* figure, additional settings are available for the F-Tile FHT PMA. The PMA is set to the default values in the PAM4 designs.

- **Pre-emphasis tap**:
  - **Post-tap 2**: Specifies the amount of pre-emphasis on the second post-tap of the transmitter buffer.
  - **Post-tap 3**: Specifies the amount of pre-emphasis on the third post-tap of the transmitter buffer.
  - **Post-tap 4**: Specifies the amount of pre-emphasis on the fourth post-tap of the transmitter buffer.
  - **Pre-tap 3**: Specifies the amount of pre-emphasis on the third pre-tap of the transmitter buffer.

Figure 20. The QSFPDD800 PAM4 Tab—PMA Setting



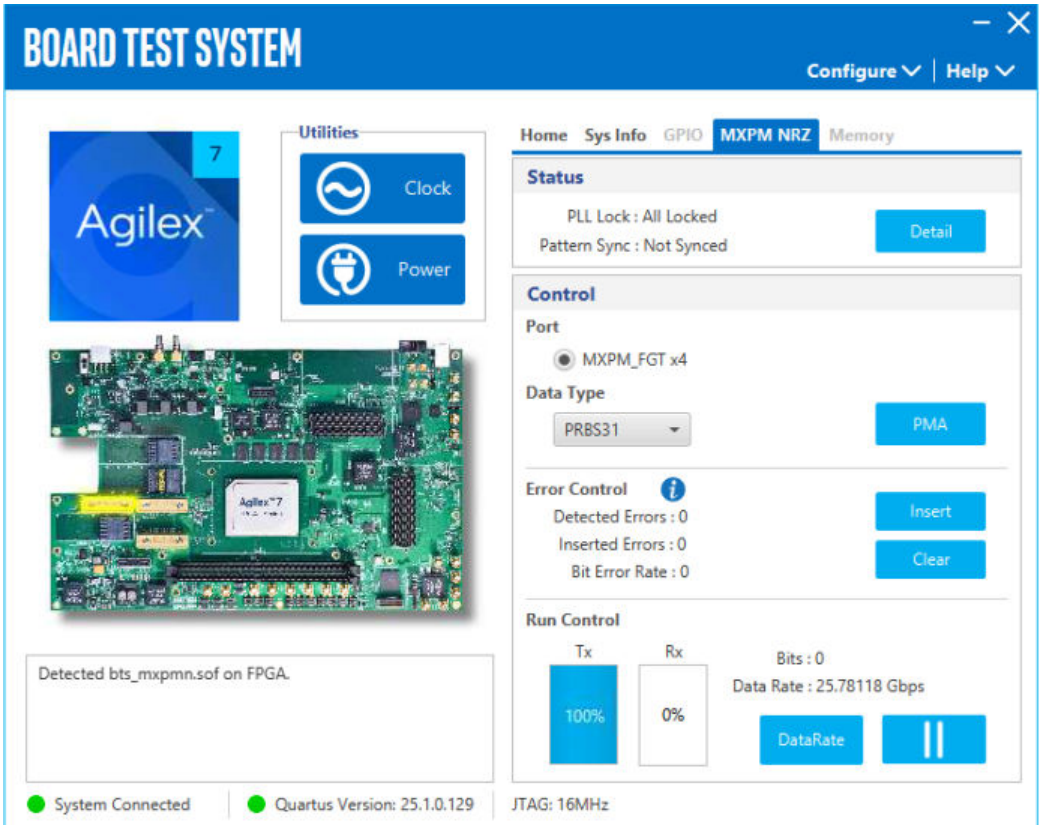
Related Information

The QSFPDD NRZ Tab on page 23

4.2.5.5. The MXPM NRZ Tab

This tab has similar control functions with the **QSFPDD NRZ** tab except for the port selection.

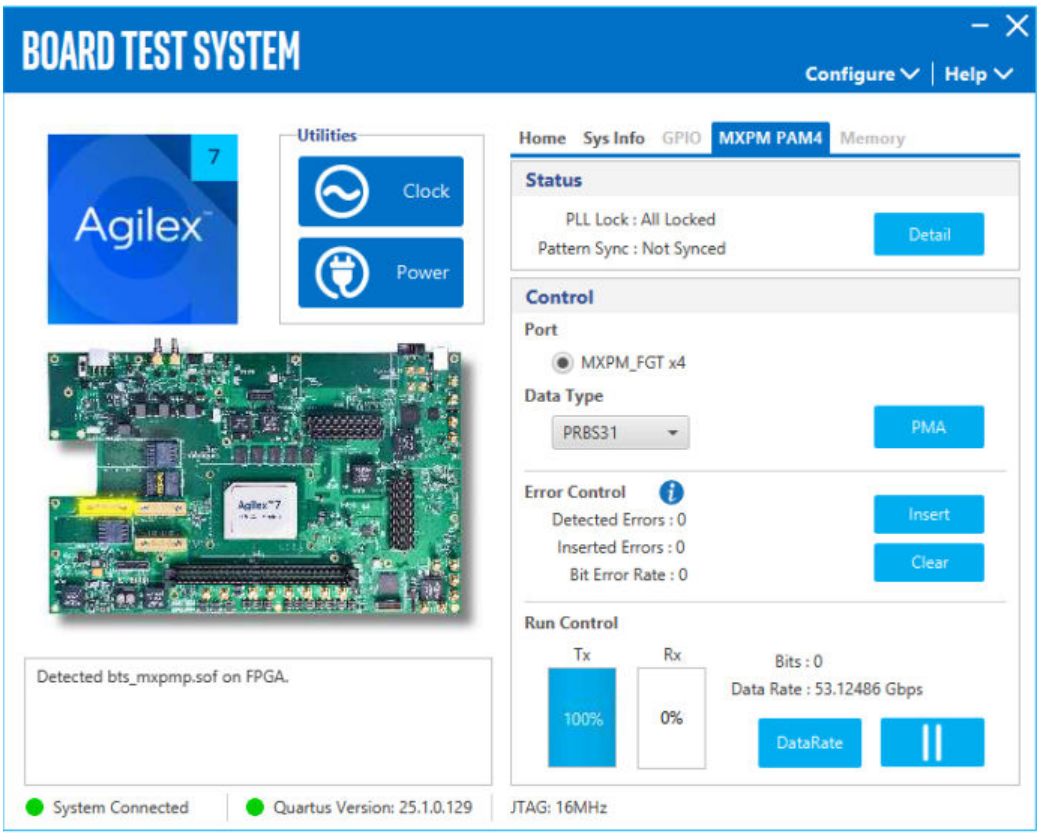
Figure 21. The MXPM NRZ Tab



4.2.5.6. The MXPM PAM4 Tab

This tab has similar control functions with the **QSFPDD NRZ** tab.

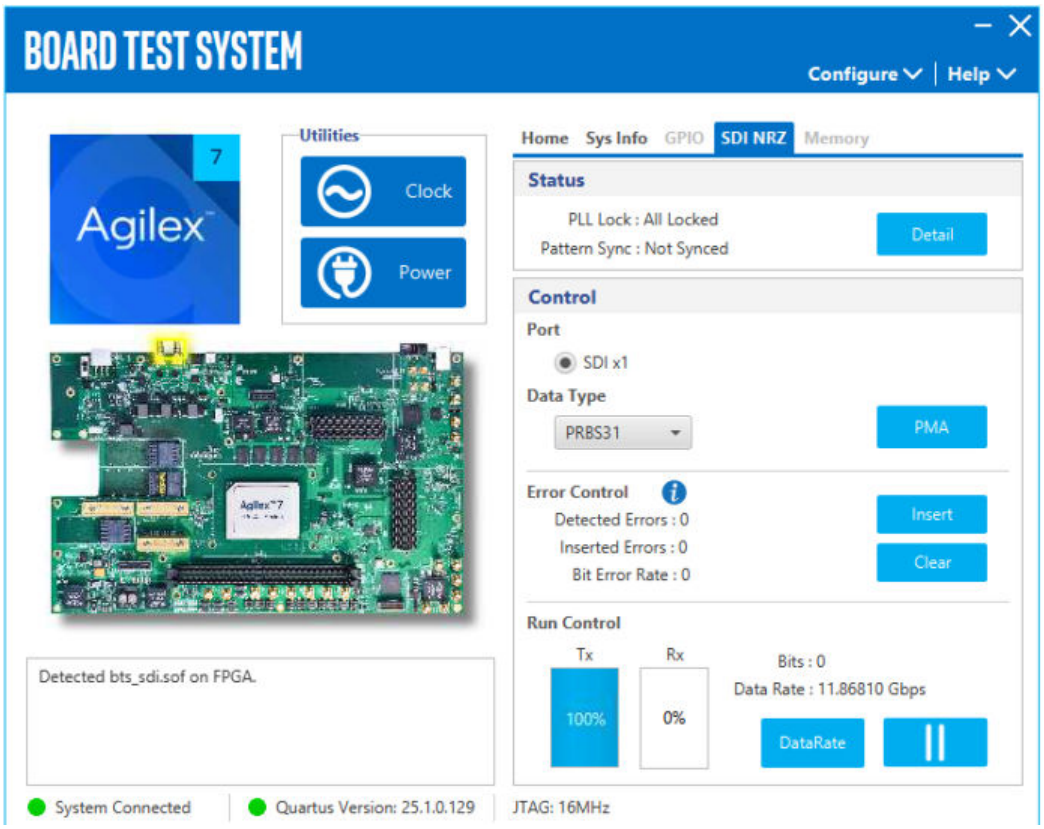
Figure 22. The MXPM PAM4 Tab



4.2.5.7. The SDI NRZ Tab

This tab has similar control functions with the **QSFPDD NRZ** tab.

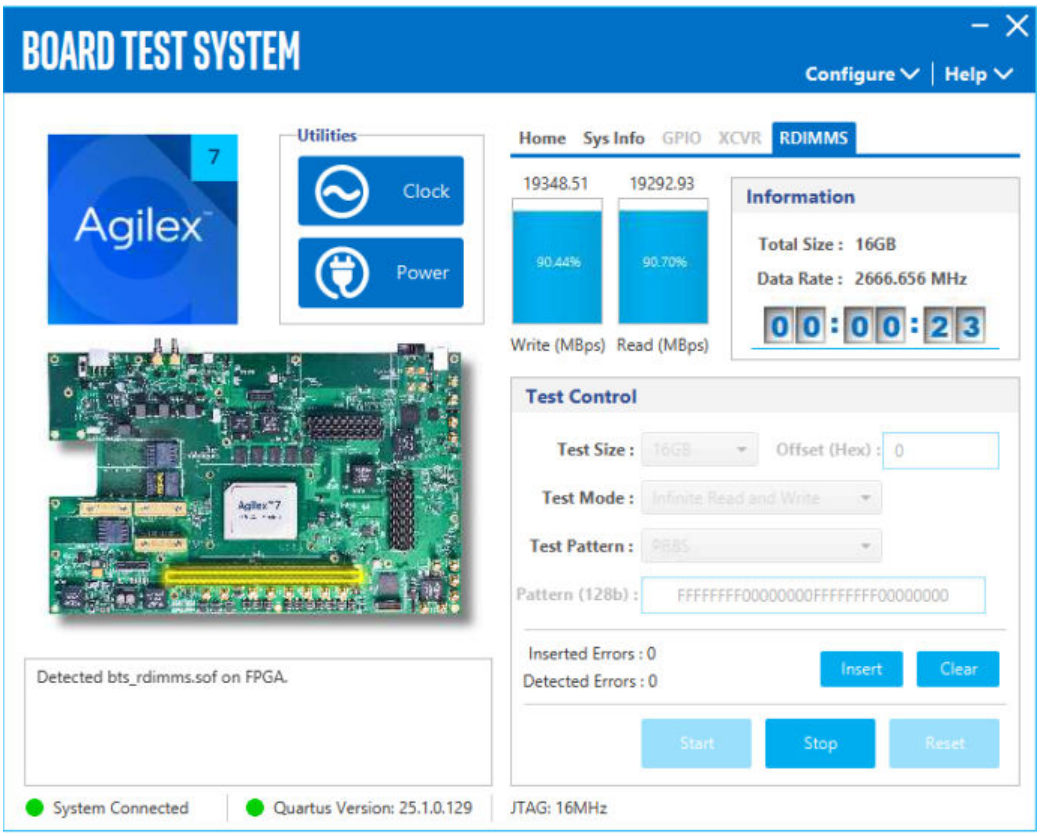
Figure 23. The SDI NRZ Tab



4.2.6. The Memory Tab

This tab allows you to read and write DDR4 DIMM-2A (DIMM-I) and DDR4 DIMM-2B (DIMM-II) memory on your board. RDIMMS only tests DIMM-II while RDIMMD tests DIMM-I and DIMM-II. Download the design through the **Configure** menu.

Figure 24. The RDIMMS Tab



The following sections describe controls on this tab.

Start

Initiates DDR4 memory transaction performance analysis.

Stop

Terminates transaction performance analysis.

Reset

Resets transaction performance analysis.

Performance Indicators

These controls display current transaction performance analysis information collected since you last clicked **Start**:

- **Write** and **Read** performance bars: Show the percentage of maximum theoretical data rate that the requested transactions are able to achieve.
- **Write (MBps) and Read (MBps)**: Show the number of bytes analyzed per second.
- **Data Bus**: 72 bits (8 bits ECC) wide, reference clock is 166.666 MHz, and the frequency is 1,333.33 MHz double data rate 2,666.66 MT/s.

Test Control

- **Test Size**: You can choose the size of the memory to test. The available options are 64 KB, 256 KB, 1 MB, 16 MB, 64 MB, 256 MB, 1 GB, 4 GB, 8 GB, and 16 GB (default).
- **Offset (Hex)**: You can define the memory start address to test.
- **Test Mode**: Infinite Read and Write (default), Single Read and Write.
- **Test Pattern**: PRBS (default), User Defined Constant, Walking '0', Walking '1'.

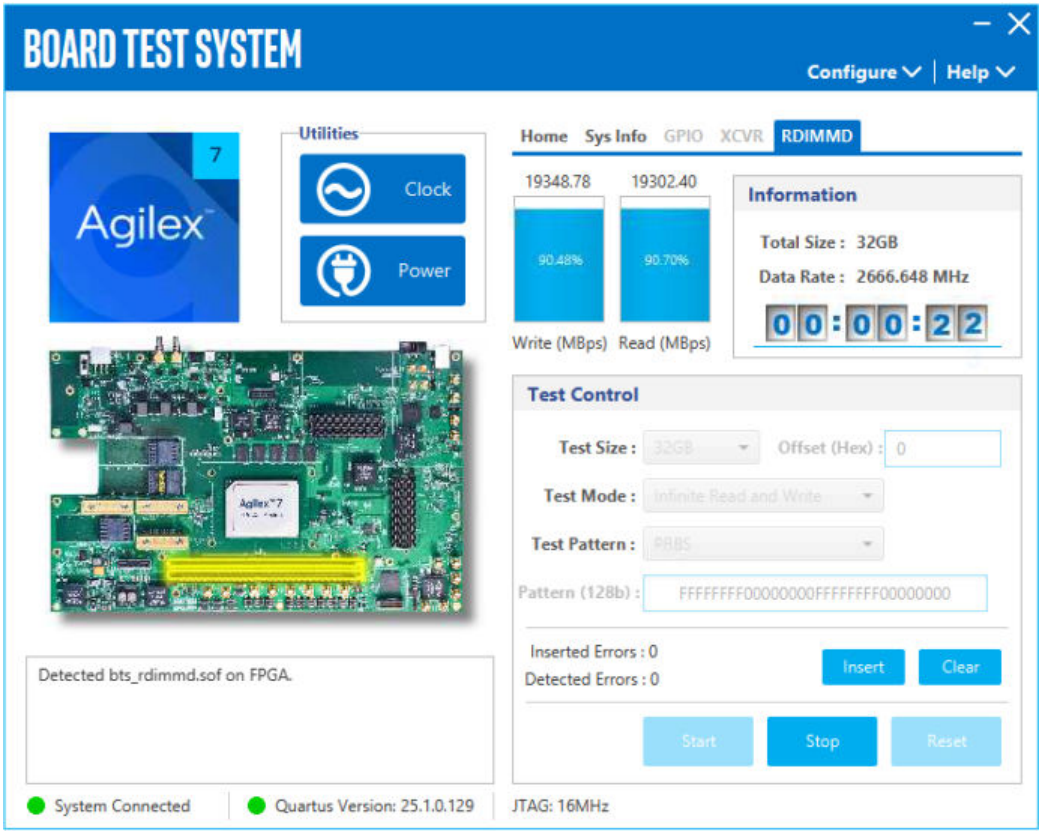
Error Control

This control displays data errors detected during analysis and allows you to insert errors:

- **Detected Errors**: Displays the number of data errors detected in the hardware.
- **Inserted Errors**: Displays the number of errors inserted into the transaction stream.
- **Insert**: Insert a one-word error into the transaction stream each time you click the button. **Insert** is only enabled during transaction performance analysis.
- **Clear**: Resets the **Detected Errors** counter and **Inserted Errors** counter to zeros.

Figure 25. The RDIMMD Tab

This tab is similar with RDIMMS. You can set **Test Size** to 32 GB with two RDIMMs.



4.3. BTS Test Areas

BTS checks for hardware fault before you can use the board. If one or more BTS test items fail, it implies either a wrong hardware setting or hardware fault on specific interface.

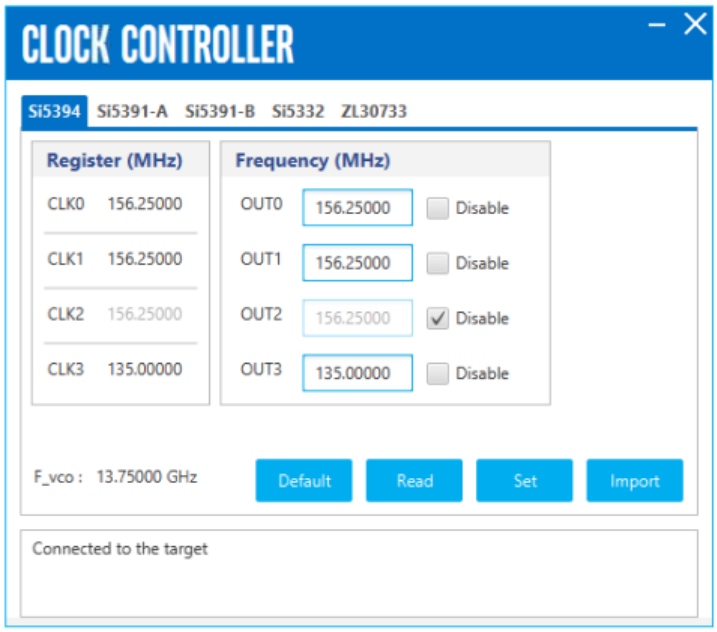
4.4. Control Onboard Clock through Clock Controller GUI

The Clock Controller communicates with the system MAX 10/Agilex 7 device through a 10-pin JTAG header `J11` or USB port `J10`. The system MAX 10 controls Si5394/Si5391-A/Si5391-B/Si5332, and Agilex 7 device controls ZL30733 through the I^2C bus.

The instructions to run the Clock Controller GUI are stated in the Running the BTS GUI section. Alternatively, you can start using the Clock Controller feature by selecting the **Clock** icon on the BTS GUI.

Note: You cannot run the stand-alone Clock Controller GUI application when the BTS or Power Monitor GUI is running at the same time. ZL30733 can be controlled only when a design in which the I^2C interface is instantiated, such as the `bts_config.sof` under the `board_test_system\image\ES folder` has been downloaded to the FPGA.

Figure 26. Si5394



The following sections describe the Clock Controller GUI buttons.

Read

Reads the current frequency setting for the oscillator associated with the active tab.

Default

Sets the frequency for the oscillator associated with the active tab back to its default value. This can also be accomplished by power cycling the board.

Set

Sets the programmable oscillator frequency for the selected clock to the value in the `OUTx` output controls for Si5394. Frequency changes might take several milliseconds to take effect. You might see glitches on the clock during this time. Altera recommends resetting the FPGA logic after changing frequencies.

Import

Si5394 has a two-time rewritable non-volatile memory (NVM). You can generate the register list from the Clockbuilder Pro tool and import it into Si5394 to update the settings of the RAM. Register changes are volatile after power cycling.

Figure 27. Si5391-A

This tab has similar control functions with Si5394.

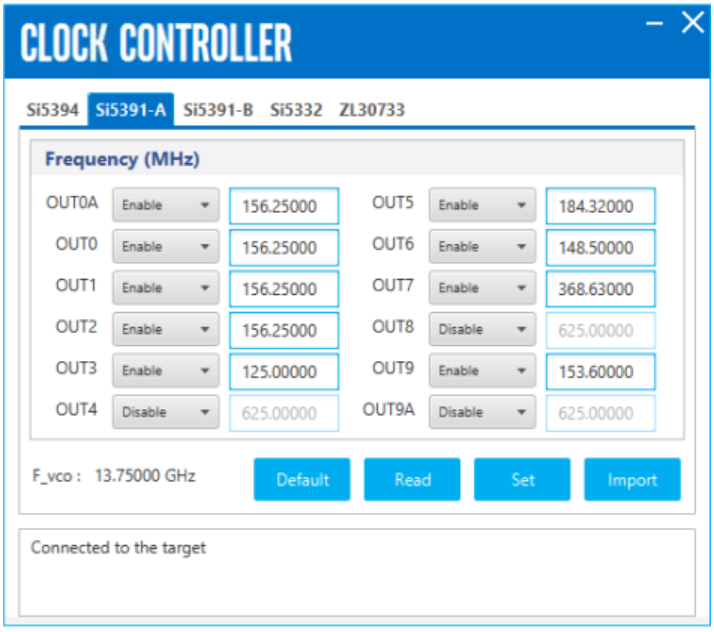


Figure 28. Si5391-B

Same with Si5391-A.

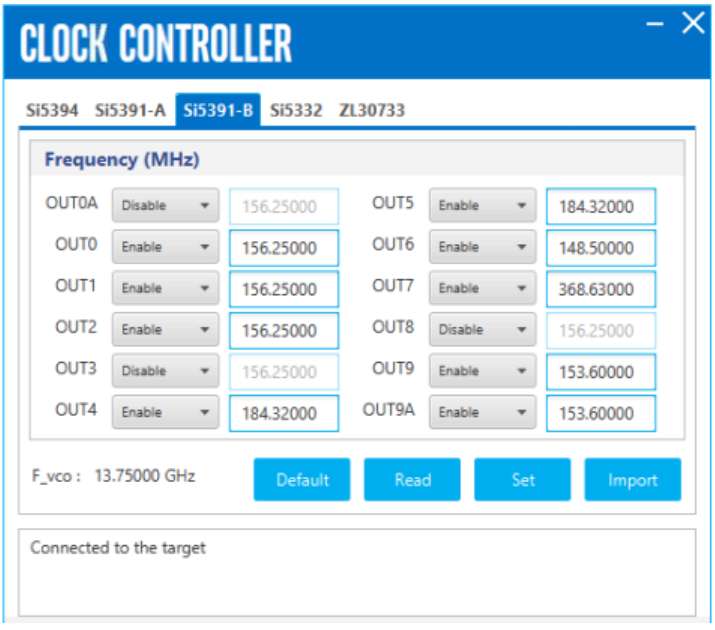


Figure 29. Si5332

This tab has similar control functions with Si5394. There is no default button for this clock but you can use Import or power cycle the board to get default clock frequencies.

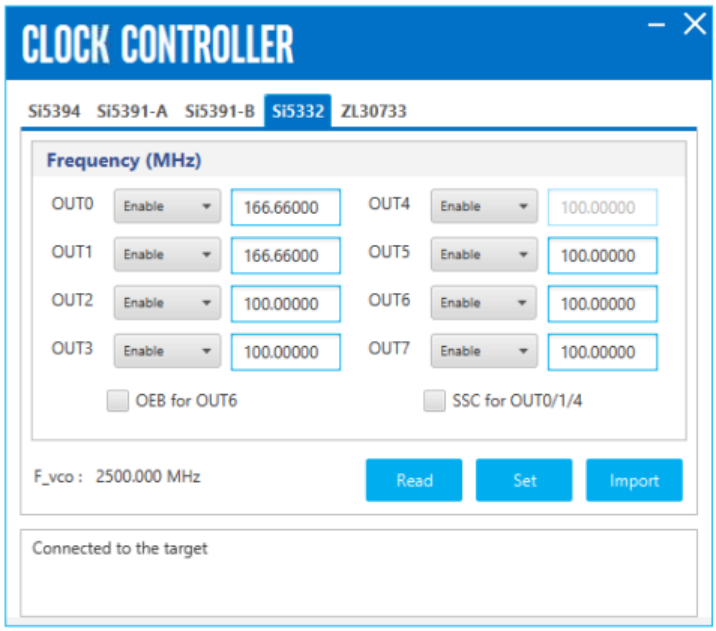
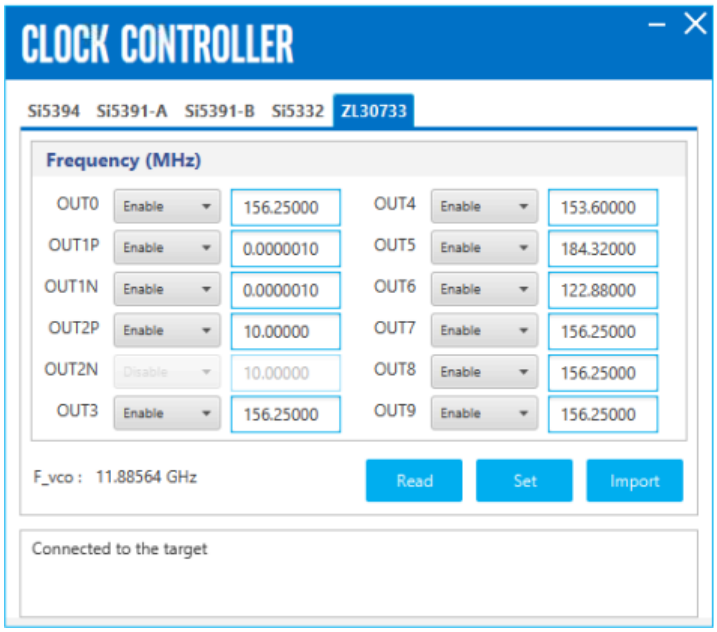


Figure 30. ZL30733

This tab has similar control functions with Si5394.



The output frequency of ZL30733 is from 0.5 Hz to 750 MHz. It can generate up to 10 differential outputs with a total of 5 output frequency families.

Import

ZL30733 has a multiple time writable non-volatile memory (NVM). You can generate the register list with the following format:

- **Register Write Command Link**: X,<register_address>, <data_bytes>, <register_address> and <data_bytes>, where prefix must be "0x" are in hexadecimal
- **Wait Command Line**: W, <time_microseconds>

Import it into ZL30733 to update the RAM settings. Register changes are volatile after power cycling. For more details, refer to ZL30733 on the Microchip website.

Related Information

Skyworks Solution
More information about the Clockbuilder Pro software.

4.5. Monitor Onboard Power Regulator through Power Monitor GUI

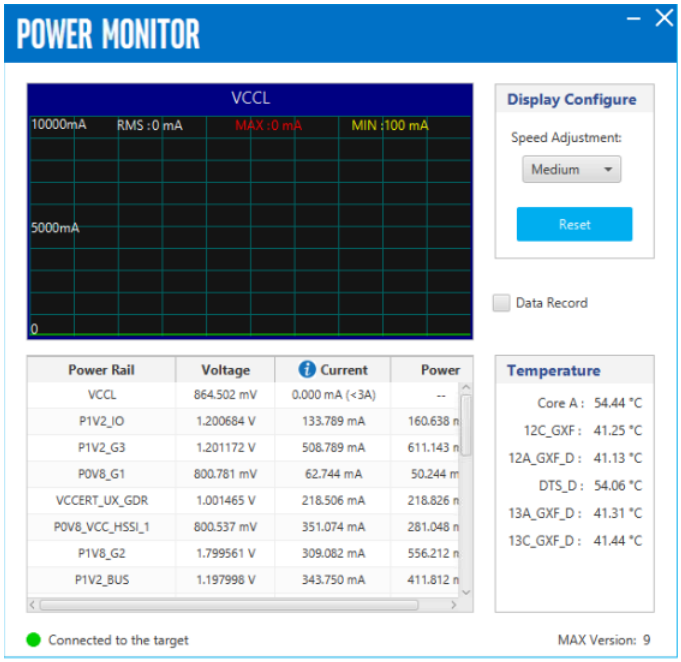
The Power Monitor GUI reports most power rails' voltage, current, and power information on the board. It also collects temperature from FPGA die, power modules, and diodes assembled on PCB.

The Power Monitor GUI communicates with the system MAX 10 through a 10-pin JTAG header `J11` or USB port `J10`. The system MAX 10 monitors and controls power regulator, temperature/voltage/current sensing chips through a 2-wire I^2C bus.

The instructions to run the Power Monitor GUI are stated in the Running the BTS GUI section. Alternatively, you can start using the Power Monitor feature by selecting the **Power** icon on the BTS GUI.

Note: You cannot run the stand-alone Power Monitor GUI when the BTS or the Clock Controller GUI is running at the same time.

Figure 31. Power Monitor GUI



The following sections describe the details of the Power Monitor GUI.

Display Configure

- **Speed Adjustment**: Adjusts the update rate of the current curve.
- **Reset**: Regenerates the graph.

Data Record

When the box is checked, the telemetry data of the selected power rail can be recorded. It saves the data into a `.csv` file in the `log` directory.

Temperature

Reads the temperature data from FPGA die internal temperature sense diodes.

4.6. Identify Test Pass or Fail based on BTS GUI Test Status

QSFPDD0/QSFPDD1

Plug the QSFPDD0/QSFPDD1 loopback module in `J27`/`J48` before you configure QSFPDD NRZ/PAM4 example build through the BTS GUI.

QSFPDD800

Plug the QSFPDD800 loopback module in `J22` and use the MXPM cable between MXPM_FHT connector `J2` and `J8` before you configure QSFPDD800 PAM4 example build through the BTS GUI.

FMCA/FMCB

Plug the FMCA/FMCB loopback module in J7/J9 before you configure FMCA/FMCB NRZ example build through the BTS GUI.

MXPM_FGT

Plug the MXPM_FGT loopback module in J28 before you configure MXPM NRZ/PAM4 example build through the BTS GUI.

DDR4 DIMM

Plug the DIMM-I/II module, which is shipped alone with this development kit, in J5. The BTS GUI only supports fabric memory interfaces, namely DDR4 DIMM-2A (DIMM-I) and DDR4 DIMM-2B (DIMM-II).

4.7. BTS Test Areas

BTS checks for hardware fault before you can use the board. If one or more BTS test items fail, it implies either a wrong hardware setting or hardware fault on specific interface.

5. Development Kit Hardware and Configuration

The Agilex 7 FPGA I-Series Transceiver-SoC Development Kit can support multiple application scenarios and configuration modes. You need to change hardware setting and/or re-program system images for these cases.

Table 5. Supported Configuration Modes

S9 [1:4]	MSEL [2:0]	Configuration Mode
ON/OFF/OFF/X	001	AS – Fast mode (Default setting)
ON/ON/OFF/X	011	AS – Normal mode
ON/ON/ON/X	111	JTAG
OFF/ON/ON/X	110	Avalon-ST x8
ON/OFF/ON/X	101	Avalon-ST x16
OFF/OFF/OFF/X	000	Avalon-ST x32

5.1. Configuring the FPGA and Accessing HPS Debug Access Port by JTAG

1. JTAG access does not rely on switch S9 settings and system image.
2. Plug the USB cable to J10 or Altera FPGA Download Cable to J11.
3. Open the Quartus Prime Programmer, system console to configuration Agilex 7 FPGA SDM, system MAX 10 and FMC JTAG nodes.
4. Open Arm* Development Studio 5* (DS-5*) Altera SoC FPGA Edition to connect to and communicate with the HPS Debug Access Port (DAP) through the same JTAG interface.

5.2. Configuring the FPGA Device Using the AS Mode (Default Mode)

1. Default S9 setting and system MAX 10 image support the active serial (AS) configuration mode.
2. Plug pre-programmed SDM QSPI flash daughter into J3.
3. Power on and observe FPGA user LED behavior.

5.3. Configuring the FPGA Device Using the Avalon-ST Mode

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1. Set S9 to Avalon-ST x32 mode first.
2. Default system MAX 10 image support the Avalon-ST x32 mode only. You should build a corresponding .POF image if you select Avalon-ST x8 or Avalon-ST x16 configuration mode.
3. Power on the board, use push button S16 to choose page, and S17 to configure the FPGA.
4. Default page depends on the setting of S10.1.

Table 6. Avalon-ST x32 LED Behavior

LED	Page0	Page1	Page2	Page3
LED_D9	ON	OFF	OFF	OFF
LED_D11	OFF	ON	OFF	ON
LED_D13	OFF	OFF	ON	ON
USER_LED0_D1	Blinking	Blinking	Blinking	Blinking
USER_LED1_D3	ON	ON	ON	ON
USER_LED2_D5	Blinking reversed	Blinking	Blinking	Blinking
USER_LED3_D7	ON	ON	ON	ON
USER_LED4_D2	ON	ON	Blinking reversed	Blinking
USER_LED5_D4	ON	Blinking reversed	Blinking	Blinking
USER_LED6_D6	ON	ON	ON	Blinking reversed
USER_LED7_D8	ON	ON	ON	ON

5.4. Daughter Cards

The Agilex 7 FPGA I-Series Transceiver-SoC Development Kit supports HPS OOBE daughter card. You can demonstrate HPS functions through these daughter cards and cables.

5.4.1. HPS Out of Box Experience (OOBE) Daughter Card

1. Plug HPS OOBE daughter card in J4.
2. To test HPS Ethernet capability, connect OOBE's RJ45 port J3 to internet.
3. To test HPS USB 2.0 capability, connect OOBE's J4 port to USB cable.
4. To debug HPS from UART terminal applications, use USB cable to connect to OOBE's J7.
5. OOBE's micro SD card is pre-programmed with GSRD and OS.
6. General I/O access is provided by push buttons and LED indicators.

6. Custom Projects for the Development Kit

6.1. Adding SmartVID Settings in Quartus Prime Settings File (QSF)

The Agilex 7 silicon assembled on this development kit enables SmartVID feature by default. Use the following settings for your Quartus Prime Settings File (`.qsf`) based on your board version:

1. Open your Quartus Prime Settings File (`.qsf`), copy and paste the following constraint scripts into the file.

   For Power Solution 2 board (DK-SI-AGI027FD, DK-SI-AGI027FC, and DK-SI-AGI027FA—Non-Enpirion*):

   ```
   set_global_assignment -name USE_PWRMGT_SCL SDM_IO0
   set_global_assignment -name USE_PWRMGT_SDA SDM_IO12
   set_global_assignment -name USE_CONF_DONE SDM_IO16
   set_global_assignment -name PWRMGT_SLAVE_DEVICE_TYPE LTC3888
   set_global_assignment -name PWRMGT_SLAVE_DEVICE0_ADDRESS 62
   set_global_assignment -name PWRMGT_VOLTAGE_OUTPUT_FORMAT "LINEAR FORMAT"
   set_global_assignment -name PWRMGT_LINEAR_FORMAT_N "-12"
   ```

   For Power Solution 1 board (DK-SI-AGI027FB and DK-SI-AGI027FES—Enpirion):

   ```
   set_global_assignment -name USE_PWRMGT_SCL SDM_IO0
   set_global_assignment -name USE_PWRMGT_SDA SDM_IO12
   set_global_assignment -name USE_CONF_DONE SDM_IO16
   set_global_assignment -name VID_OPERATION_MODE "PMBUS MASTER"
   set_global_assignment -name PWRMGT_BUS_SPEED_MODE "100 KHZ"
   set_global_assignment -name PWRMGT_SLAVE_DEVICE_TYPE ED8401
   set_global_assignment -name NUMBER_OF_SLAVE_DEVICE 1
   set_global_assignment -name PWRMGT_SLAVE_DEVICE0_ADDRESS 62
   set_global_assignment -name PWRMGT_VOLTAGE_OUTPUT_FORMAT "LINEAR FORMAT"
   set_global_assignment -name PWRMGT_LINEAR_FORMAT_N "-13"
   set_global_assignment -name PWRMGT_TRANSLATED_VOLTAGE_VALUE_UNIT VOLTS
   ```

2. Ensure there are no other similar settings with different values.

6.2. Golden Top

You can use the Golden Top project as the starting point for your designs. It comes loaded with constraints, pin locations, defined I/O standard, direction, and general termination. The DDR4 pin termination settings are not included. Refer to the bts_emif example designs for details.

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7. Document Revision History for the Agilex 7 FPGA I-Series Transceiver-SoC Development Kit User Guide

Document Version	Changes
2026.05.22	• Updated the *Overview* chapter: ☐ Updated Figure: *Agilex 7 FPGA I-Series Transceiver-SoC Development Kit (Power Solution 2 Board)—Top View*. ☐ Updated Figure: *Agilex 7 FPGA I-Series Transceiver-SoC Development Kit Block Diagram*. • Updated the *Getting Started chapter*: ☐ Updated *Software and Driver Installation*. ☐ Updated *Installing the Altera SoC FPGA Embedded Development Suite (EDS)*. ☐ Updated the figure and the table information in *Installing the Development Kit*. ☐ Removed topic—*Installing the Intel® FPGA Download Cable Driver*. • Updated the *Board Test System* chapter: ☐ Added a note to *Set Up BTS GUI Running Environment*. ☐ Updated the installation steps for Windows and Linux systems in *Installing OpenJDK and OpenJFK*. ☐ Updated the instructions for Linux system in *Running the BTS GUI*. • Updated the *Development Kit Components* appendix chapter: ☐ Updated Table: *Default Clock Frequency*. ☐ Updated Figure: *Clock Tree*. ☐ Added the following signals to Table: *UB2/PWR MAX 10*: • `FTDI_UART_TXD` • `FTDI_UART_RXD` • Updated the *Custom Projects for the Development Kit*: ☐ Retitled *Adding SmartVID Settings in the Quartus QSF File* to *Adding SmartVID Settings in Quartus Prime Settings File (QSF)*. ☐ Updated the constraint script for Power Solution 2 board in *Adding SmartVID Settings in Quartus Prime Settings File (QSF)*. ☐ Updated *Golden Top*.
2026.01.05	• Added information about ordering code "DK-SI-AGI027D" to the following topics: ☐ *Overview* ☐ *Board Test System* ☐ *Adding SmartVID Settings in the Quartus Prime QSF File* ☐ *Power* ☐ *Communication Interfaces* • Added new topic—*Handling the DIP Switches*.
2025.11.05	• Updated Figure: *Agilex 7 FPGA I-Series Transceiver-SoC Development Kit Block Diagram* for clarity. • Updated the document for the latest branding standards. • Made editorial edits throughout the document.

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Document Version	Changes
2025.02.28	• Updated the *Overview* chapter: ☐ Updated the following topics: • *Recommended Operating Conditions* • Updated the *Getting Started* chapter: ☐ Added new topics: • *Before You Begin* • *Handling the Board* • *Installing the Quartus Prime Pro Edition Software* • *Installing the Intel SoC EDS* • *Installing the Development Kit* • *Installing the Altera FPGA Download Cable Driver* ☐ Updated and retitled topic *Quartus Prime Software and Driver Installation* to *Software and Driver Installation*. ☐ Removed the *Design Examples* topic. • Updated and retitled chapter *Power Up the Development Kit* to *Development Kit Setup*: ☐ Retitled topic *Power Up* to *Powering Up the Development Kit*. ☐ Retitled topic *Restore Board System MAX 10 with Default Factory Image* to *Restoring Board System MAX 10 with Default Factory Image*. • Updated the *Board Test System* chapter: ☐ Retitled topic *Download OpenJDK* to *Downloading OpenJDK*. ☐ Retitled topic *Download OpenJFX* to *Downloading OpenJFX*. ☐ Retitled topic *Download OpenJDK and OpenJFX* to *Downloading OpenJDK and OpenJFX*. ☐ Retitled topic *Install Quartus Prime Software* to *Setting Up the Quartus Prime Software for BTS Operation*. ☐ Retitled topic *Run BTS GUI* to *Running the BTS GUI*. ☐ Updated the *Control On-Board Clock through Clock Controller GUI* topic. • Added new appendix chapter—*Developer Resources*. • Restructured the document to improve clarity and for ease of reference. • Updated the document for the latest branding standards.
2024.05.31	• Updated the description of Development Kit Version in Table: *Ordering Information*.
2023.04.10	• Updated Ordering Information in *Overview* section. • Added Figure: *Agilex 7 FPGA I-Series Transceiver-SoC Development Kit Board Image - DK-SI-AGI027FB and DK-SI-AGI027FES (Enpirion)* and Figure: *Agilex 7 FPGA I-Series Transceiver-SoC Development Kit Board Image - DK-SI-AGI027FA and DK-SI-AGI027FC (Not Enpirion)* in *Overview* section. • Updated list in *Box Contents* section. • Updated the FPGA maximum power supported by active heatsink/fan in *Recommended Operating Conditions*. • Updated *Restore Board QSPI Flash with the Default Factory Image* section. • Added *Control On-board Clock through Clock Controller GUI* section. • Added *Control On-board Power Regulator through Power Monitor GUI* section. • Added Post-tap 2 setting in *The QSFPDD800 PAM4 Tab* section. • Added *The SDI Tab* section. • Added description for Reset button in *The Memory Tab* section. • Updated description for *Custom Projects for the Development Kit*. • Updated Figure: *BTS GUI*. • Updated Figure: *The Sys Info Tab*. • Updated Figure: *The Configure Menu*. • Updated Figure: *The Sys Info Tab*. • Updated Figure: *The GPIO Tab*. • Updated Figure: *The QSFPDD NRZ Tab*. • Updated Figure: *QSFPDD NRZ – PLL and Pattern Status*. • Updated Figure: *QSFPDD NRZ - PMA Setting*.

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Document Version	Changes
	• Updated Figure: *QSFPDD NRZ - Data Rate*. • Updated Figure: *The QSFPDD PAM4 Tab*. • Updated Figure: *The FMCA NRZ Tab*. • Updated Figure: *The FMCB NRZ Tab*. • Updated Figure: *The QSFPDD800 PAM4 Tab*. • Updated Figure: *The MXPM NRZ Tab*. • Updated Figure: *The MXPM PAM4 Tab*. • Updated Figure: *The RDIMMS Tab*. • Updated Figure: *The RDIMMD Tab*. • Updated Figure: *ZL30733*. • Updated Figure: *Power Monitor GUI*. • Added constraint scripts for DK-SI-AGI027FA and DK-SI-AGI027FC, DK-SI-AGI027FB and DK-SI-AGI027FES in *Add SmartVID Settings in the Intel Quartus Prime QSF File* section. • Added Figure: *Power Tree (Not Enpirion) DK-SI-AGI027FA and DK-SI-AGI027FC*. • Added Figure: *I^2C Serial Bus (Not Intel Enpirion) DK-SI-AGI027FA and DK-SI-AGI027FC*. • Updated product family name to "Intel Agilex® 7".
2023.03.02	Updated the MXMP breakout cable and MCIO cable part number details in Table: *Connectors and Cables*.
2022.09.30	• Added UKCA in the *Safety and Regulatory Information* and *Compliance Information* sections. • Added the *Lithium Ion Battery Warnings* section. • Updated Table: *Ordering Information*. • Updated the *Perform Board Restore* section. • Updated step 3 in the *Restore Board QSPI Flash with Default Factory Image* section. • Updated the *Control On-Board Clock through Clock Controller GUI* section. • Updated step 1 and step 3 in the *Configure FPGA and Access HPS Debug Access Port by JTAG* section. • Updated the *System Management* section. • Updated the *Power Cord Requirements* section. • Updated the *Safety Cautions* section.
2022.04.07	Initial release.

A. Development Kit Components

A.1. System Management

Two MAX 10 FPGAs (10M16SCU324C8G) are used for system management. The system MAX 10 acts as a system controller. It handles the FPGA Avalon streaming configuration, I^2C bus access, fan speed control, and system reset functions. The UB2/PWR MAX 10 acts as the Power Manager and onboard JTAG controller. Refer to the following description for each function:

- **Power management**: Control systems and FPGA power-up and optional power-down sequence (PDS), supervise power regulators/switches status and manage power faults, supervise temperature analog-to-digital converter (ADC) interrupt signals and manage temperature faults.
- **JTAG controller**: Manage the JTAG chain topology, JTAG master source, and JTAG slaves using S19/S20.

Table 7. JTAG Master Sources

Schematic Signal Name	Description
EXT_JTAG_TCK/TDO/TMS/TDI	JTAG header J11 for Altera FPGA Download Cable
FX2_Dp/n	Input port J10 for onboard Altera download circuit
HPS_GPIO[32:35]	Mictor 38-pin header on the OOBE daughter card

Agilex 7 HPS JTAG can be accessed from either SDM dedicated JTAG pins or HPS dedicated I/Os. When it is accessed from SDM JTAG pins (EXT_JTAG_TCK/TDO/TMS/TDI), SDM is chained with HPS inside FPGA. When it is accessed from HPS dedicated I/Os (HPS_GPIO[32:35]), HPS is chained externally by PCB traces.

Table 8. JTAG Chain Topology Settings

Mode	S20 [4:2]	S19[4][3][2][1] ON: Bypass from chain OFF: Enable in chain	Function
000	ON/ON/ON (Default)	S19.1 (SDM+HPS) S19.2 (System MAX 10) S19.3 (FMC_B) S19.4 (FMC_A)	Mode 1: Onboard Altera download circuit act as the only JTAG Master. Chained HPS with SDM nodes **internally**.

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Mode	S20 [4:2]	S19[4][3][2][1] ON: Bypass from chain OFF: Enable in chain	Function
			Mode 3: External Altera FPGA Download Cable act as the only JTAG Master. Chained HPS with SDM nodes **internally**.
001	ON/ON/OFF	SDM is always enabled in the JTAG chain. `S19.1` (HPS) `S19.2` (System MAX 10) `S19.3` (FMC_B) `S19.4` (FMC_A)	Mode 2: Onboard Altera download circuit act as the only JTAG Master. Chained HPS with SDM nodes **externally**. Mode 4: External Altera FPGA Download Cable act as the only JTAG Master. Chained HPS with SDM node **externally**.
100	OFF/ON/ON	`S19.1` (SDM) `S19.2` (System MAX 10) `S19.3` (FMC_B) `S19.4` (FMC_A)	Mode 7: Both onboard Altera download circuit and OOBE act as JTAG Masters. Separated HPS and SDM JTAG **chains, OOBE only drive HPS**. Mode 8: Both external Altera FPGA Download Cable and OOBE JTAG act as JTAG Masters. Separated HPS and SDM JTAG **chains, OOBE only drive HPS**.
Others	N/A	N/A	Reserved

A.2. Power

Figure 32. Power Tree (Power Solution 2 Board—non-Enpirion)

This figure applies to DK-SI-AGI027FD, DK-SI-AGI027FC, and DK-SI-AGI027FA.

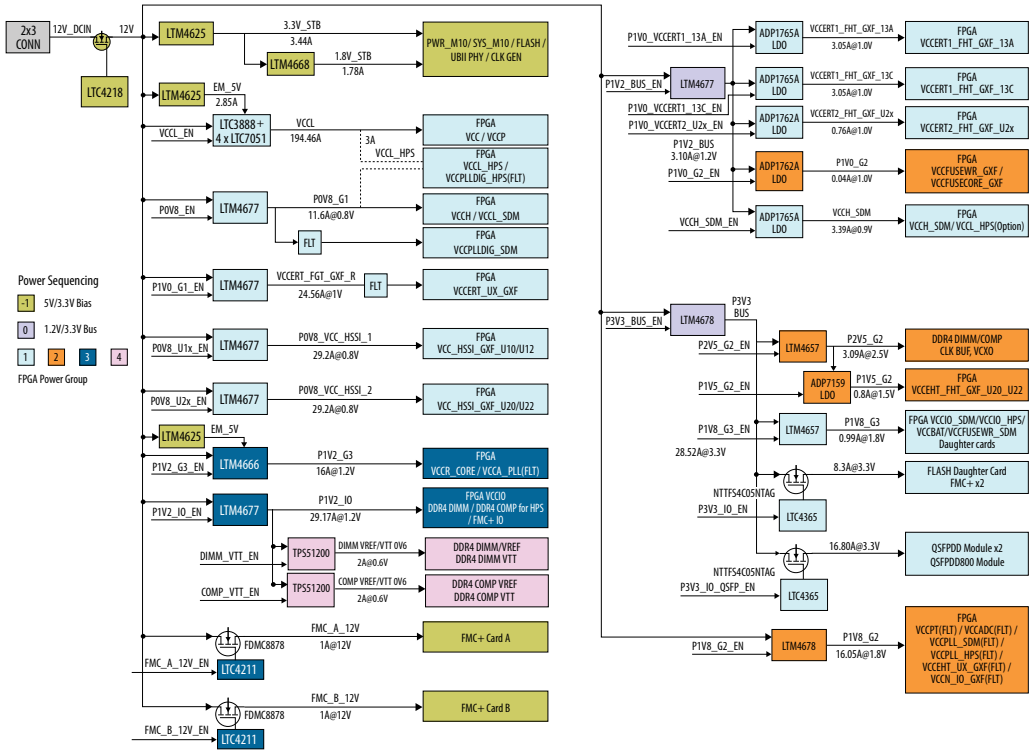


Figure 33. Power Tree (Power Solution 1 Board—Enpirion)

This figure applies to DK-SI-AGI027FB and DK-SI-AGI027FES.

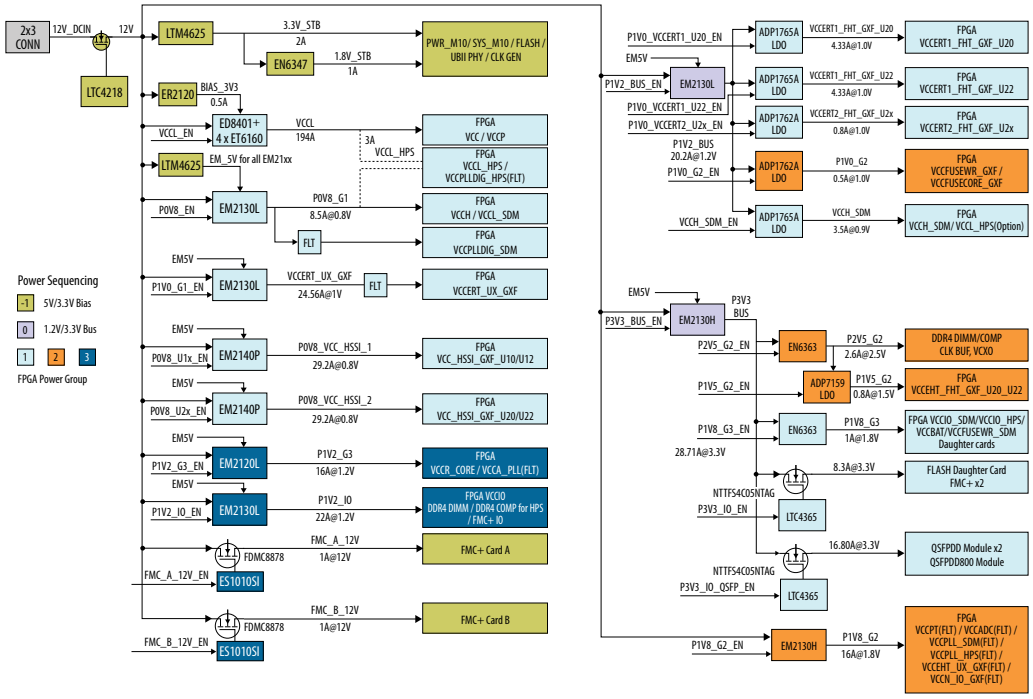
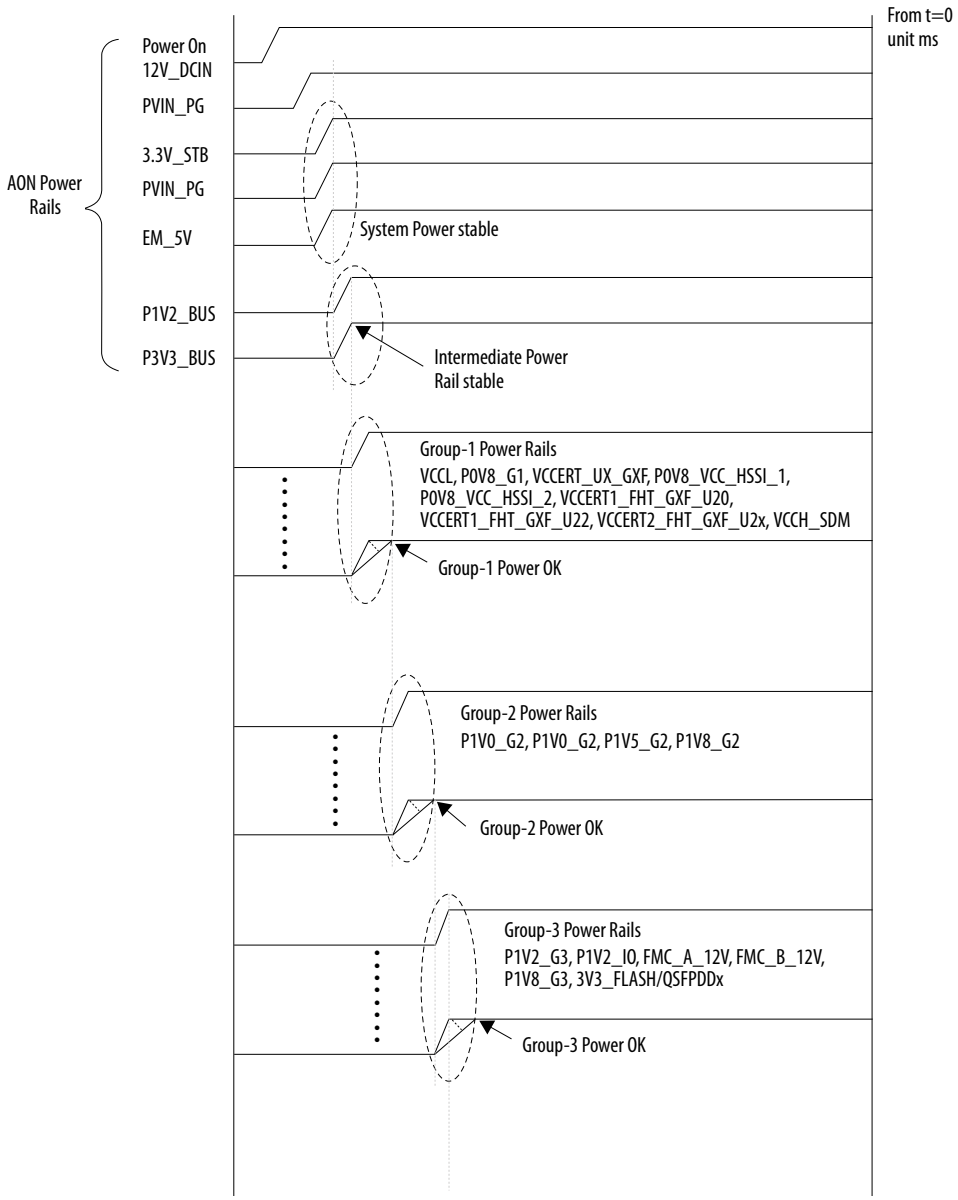


Figure 34. Power Sequence



Onboard hot-plug circuit shuts down all power rails when the total power is over 360 W (30 A).

UB2/PWR MAX 10 shuts down significant power rails when one or more good power indicators is low due to a power fault.

UB2/PWR MAX 10 also shuts down significant power rails when temperature cross the acceptable range.

A.3. Clocks

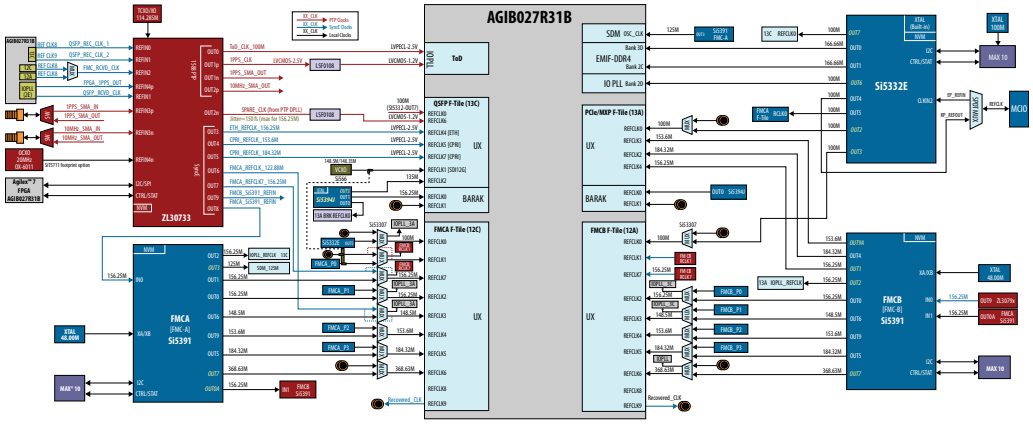
Table 9. Default Clock Frequency

Schematic Signal Name	Default Frequency
DDR4_HPS_REFCLKp/n	166.6M
DDR4_DIMM_2_REFCLKp/n	166.6M
Si5332_OUT2_P/N	100M
Si5332_OUT3_P/N	100M
PCIE_100M_REF_RP_AP/N	100M
Si5332_OUT5_P/N	100M
CLK_GPIO_P_4/N_4	100M
Si5332_OUT7_P/N	100M
Si5391_A_OUT_P0A/N0A	156.25M
Si5391_A_OUT_P0/N0	156.25M
Si5391_A_OUT_P1/N1	156.25M
Si5391_A_OUT_P5/N5	184.32M
Si5391_A_OUT_P6/N6	148.5M
Si5391_A_OUT_P7/N7	368.63M
Si5391_A_OUT_P9/N9	153.6M
Si5391_B_OUT_P0/N0	156.25M
CLK_13A_FGT_REFCLK4_P/N	156.25M
CLK_13A_FGT_REFCLK2_P/N	184.32M
Si5391_B_OUT_P5/N	184.32M
Si5391_B_OUT_P6/N6	148.5M
Si5391_B_OUT_P7/N7	368.63M
Si5391_B_OUT_P9/N9	153.6M
CLK_13A_FGT_REFCLK3_P/N	153.6M
CLK_FHT_13A_P_0/N_0	156.25M
CLK_FHT_13C_P_0/N_0	156.25M
CLK_FGT_13C_REFCLK_2P/N	135M
CLK_A_12C_FGT_P_0/N_0	100M
CLK_3A_GPIO_P_0/N_0	100M
CLK_A_12C_FGT_P_3/N_3	148.5M
CLK_3A_GPIO_P_2/N_2	148.5M
CLK_A_12C_FGT_P_2/N_2	156.25M

continued...

Schematic Signal Name	Default Frequency
CLK_3A_GPIO_P_1/N_1	156.25M
CLK_A_12C_FGT_P_4/N_4	156.25M
CLK_A_12C_FGT_P_6/N_6	368.63M
CLK_A_12C_FGT_P_5/N_5	184.32M
CLK_B_12A_FGT_P_0/N_0	100M
CLK_13A_FGT_REFCLK0_P/N	100M
CLK_B_12A_FGT_P_2/N_2	156.25M
CLK_3C_GPIO_P_1/N_1	156.25M
CLK_B_12A_FGT_P_3/N_3	148.5M
CLK_3C_GPIO_P_0/N_0	148.5M
CLK_B_12A_FGT_P_4	156.25M
CLK_B_12A_FGT_P_5	184.32M
CLK_B_12A_FGT_P_6	368.63M
ToD_CLK_156.25M_P/N	156.25M
ETH_REFCLK_156.25M_P/N	156.25M
CPRI_REFCLK_153.6M_P/N	153.6M
CPRI_REFCLK_184.32M_P/N	184.32M
FMCA_REFCLK_122.88M_P/N	122.88M
FMCA_REFCLK_156.25M_P/N	156.25M
Si5391_A_156.25M_REFIN_P/N	156.25M
Si5391_B_156.25M_REFIN_P/N	156.25M
CLK_3C_GPIO_P_1/N_1	156.25M
CLK_B_12A_FGT_P_3/N_3	148.5M
CLK_3C_GPIO_P_0/N_0	148.5M
1PPS_FPGA_CLK	1PPS
1PPS_SMA_OUT	1PPS
10MHz_SMA_OUT	10M
ZL_SPARE_CLK	100M

Figure 35. Clock Tree



A.4. General Input/Output

Table 10. MAX 10 and FPGA

Schematic Signal Name	Description
F_GPIO0	The value of filtered user_pb[0]
F_GPIO1	The value of filtered user_pb[1]
F_GPIO2	MCIO_PERST in RP mode
F_GPIO3	FMC_A_PERST in RP mode
F_GPIO4	FMC_B_PERST in RP mode
F_GPIO5	Status of SYS_SW3 from MAX 10
F_GPIO6	Reserved
F_GPIO7	Reserved
F_GPIO8	Reserved
F_GPIO9	Reserved
F_GPIO10	Reserved
F_GPIO11	Reserved
F_GPIO12	Reserved

Table 11. System MAX 10

Schematic Signal Name	Description
SYS_LED0/D9	PGM_LED0 for Avalon-ST configuration
SYS_LED1/D11	PGM_LED1 for Avalon-ST configuration
SYS_LED2/D13	PGM_LED2 for Avalon-ST configuration
SYS_LED3/D15	MAX_ERROR for Avalon-ST configuration

continued...

Schematic Signal Name	Description
SYS_LED4/D10	MAX_LOAD for Avalon-ST configuration
SYS_LED5/D12	MAX_CONF_DONE for Avalon-ST configuration
SYS_LED6/D14	Reserved
SYS_LED7/D16	Reserved
SYS_PB0/S11	MAX_RESETn
SYS_PB1/S12	FPGA_RESETn
SYS_PB2/S13	HPS_COLD_RESETn
SYS_PB3/S14	Power recycle
SYS_PB4/S16	PGM_SEL for Avalon-ST configuration
SYS_PB5/S17	PGM_CFG for Avalon-ST configuration
clk_i2c_en0	Before power ok: 0 After power ok: S_control_gui[2], =1 by default
clk_i2c_en1	Stuck at ground (GND)
VCCL_I2C_EN	Before power ok: 0 After power ok: S_control_gui[1], =1 by default
dimm_io_en	Before power ok: 0 After power ok: 1
zl_i2c_en	Before power ok: 0 After power ok: S_control_gui[3], =0 by default
si5394_rstn	When input globe resetn active: 0 When input globe resetn release: S_control_gui[13], =1 by default
si5394_oe_n	Before power ok: 1 After power ok: S_control_gui[14], =0 by default
si5391_rstn	When input globe resetn active: 0 When input globe resetn release: S_control_gui[9], =1 by default
si5394_lol	Store in status_gui bit3
si5394_int_n	Store in status_gui bit2
si5391_oen	Before power ok: 1 After power ok: S_control_gui[10], =0 by default
si5391_b_rstn	When input globe resetn active: 0 When input globe resetn release: S_control_gui[11], =1 by default
si5391_b_oen	Before power ok: 1 After power ok: S_control_gui[12], =0 by default
usb2_resetn	Before power ok: 0 After power ok: S_control_gui[15], =1 by default
usb_mux_reset	Before power ok: 1 After power ok: S_control_gui[16], =0 by default

continued...

Schematic Signal Name	Description
mux_sel0	system_info_slv_data_write_0[0]=1 : controlled by system_info_slv_data_write_1[0] system_info_slv_data_write_0[0]=0 : controlled by MUX_DIP_SW10
mux_sel1	system_info_slv_data_write_0[1]=1 : controlled by system_info_slv_data_write_1[1] system_info_slv_data_write_0[1]=0 : controlled by SYS_SW1
mux_sel2	system_info_slv_data_write_0[2]=1 : controlled by system_info_slv_data_write_1[2] system_info_slv_data_write_0[2]=0 : controlled by MUX_DIP_SW0
mux_sel3	system_info_slv_data_write_0[3]=1 : controlled by system_info_slv_data_write_1[3] system_info_slv_data_write_0[3]=0 : controlled by MUX_DIP_SW1
mux_sel4	system_info_slv_data_write_0[4]=1 : controlled by system_info_slv_data_write_1[4] system_info_slv_data_write_0[4]=0 : controlled by MUX_DIP_SW2
mux_sel5	system_info_slv_data_write_0[5]=1 : controlled by system_info_slv_data_write_1[5] system_info_slv_data_write_0[5]=0 : controlled by MUX_DIP_SW3
mux_sel6	system_info_slv_data_write_0[6]=1 : controlled by system_info_slv_data_write_1[6] system_info_slv_data_write_0[6]=0 : controlled by MUX_DIP_SW9
mux_sel7	system_info_slv_data_write_0[7]=1 : controlled by system_info_slv_data_write_1[7] system_info_slv_data_write_0[7]=0 : controlled by SYS_SW2
mux_sel8	system_info_slv_data_write_0[8]=1 : controlled by system_info_slv_data_write_1[8] system_info_slv_data_write_0[8]=0 : controlled by MUX_DIP_SW11
mux_sel9	system_info_slv_data_write_0[9]=1 : controlled by system_info_slv_data_write_1[9] system_info_slv_data_write_0[9]=0 : controlled by SYS_SW3
mux_sel10	system_info_slv_data_write_0[10]=1 : controlled by system_info_slv_data_write_1[10] system_info_slv_data_write_0[10]=0 : controlled by MUX_DIP_SW4
mux_sel11	system_info_slv_data_write_0[11]=1 : controlled by system_info_slv_data_write_1[11] system_info_slv_data_write_0[11]=0 : controlled by MUX_DIP_SW5

continued...

Schematic Signal Name	Description
mux_sel12	system_info_slv_data_write_0[12]=1 : controlled by system_info_slv_data_write_1[12] system_info_slv_data_write_0[12]=0 : controlled by MUX_DIP_SW6
mux_sel13	system_info_slv_data_write_0[13]=1 : controlled by system_info_slv_data_write_1[13] system_info_slv_data_write_0[13]=0 : controlled by MUX_DIP_SW8
mux_sel14	system_info_slv_data_write_0[14]=1 : controlled by system_info_slv_data_write_1[14] system_info_slv_data_write_0[14]=0 : controlled by MUX_DIP_SW7
mux_sel_zl	system_info_slv_data_write_0[15]=1 : controlled by system_info_slv_data_write_1[15] system_info_slv_data_write_0[15]=0 : controlled by SYS_SW4
mcio_clk_enn	Before power ok: 1 After power ok: S_control_gui[7], =0 by default
mcio_clk_sel_epn	system_info_slv_data_write_0[16]=1 : controlled by S_control_gui[8], =0 by default system_info_slv_data_write_0[16]=0 : • =0 when AGIB027R31B is Endpoint • =1 when AGIB027R31B is Root port
si5332_1_in[1:0]	S_control_gui[5:4], =00 by default
si5332_1_in[2]	1'bZ
PTP_CLK_RST_n	Before power ok: 0 After power ok: S_control_gui[6], =1 by default
PTP_CLK_LOL	Store in status_gui bit1
gpio0_ac0_zl_intn	Store in status_gui bit0

Table 12. UB2/PWR MAX 10

Schematic Signal Name	Description
FPGA_POK_LED	FPGA Power Good
SYS_PWR_RSV0	Reserved GPIO between the system MAX 10 and the power MAX 10. Used as I^2C clock.
SYS_PWR_RSV1	Reserved GPIO between the system MAX 10 and the power MAX 10. Used as I^2C data.
SYS_PWR_RSV2	Reserved GPIO between the system MAX 10 and the power MAX 10. It is the status of SYS_PB3.
SYS_PWR_RSV3	Reserved GPIO between the system MAX 10 and the power MAX 10.
FTDI_UART_TXD	FTDI UART transmit signal
FTDI_UART_RXD	FTDI UART receive signal

A.5. Memory Interfaces

FPGA Dedicated External Memory Interface (2DPC DDR4)

Agilex 7 FPGA I-Series Transceiver-SoC Development Kit supports 32 GB 2DPC DDR4 with ECC support (x72). Mechanically, the development kit provides two dual-in memory module slots for the same. This development kit also supports 16 GB 1DPC DDR4 with ECC support (x72). Mechanically, install one RDIMM module on J5.

FPGA and HPS Shared External Memory Interface (DDR4)

The DDR4 component interface is a 72 bit, single rank configuration based on x16 component. It runs at 2,666 Mbps. MT40A1G16RC-062E:B from Micron is soldered down on the development kit. Both Agilex 7 FPGA fabric and HPS can access this external memory interface. However, they cannot be accessed at the same time.

A.6. Communication Interfaces

MCIO Port

The MCIO slot is a PCIe 4.0 x4 port which fans out from Agilex 7 FPGA I-Series F-Tile. This port is designed to meet the standard MCIO pinout. The system MAX 10 acts as the board management controller (BMC) of the development kit. It manages power-up reset for both PCIe root port and PCIe endpoint. PCIE_PERSTn_A signal can act as output and input respectively.

Table 13. MCIO Port

Schematic Signal Name	Description
PCIE_PERSTn_A	PCIe endpoint/root port reset
PCIE_ALERTn_A	PCIe Alert
PCIE_100M_REF_AP/AN	PCIe reference clock
PCIE_SCL_A/SDA_A	PCIe I^2C bus
PCIE_TX_P/N[0:3]	Transceiver TX
PCIE_RX_P/N[0:3]	Transceiver RX

MCIO x4 Connector

The recommended MCIO cable to use with MCIO connector (Uxx) is NOT included as part of the development kit and must be acquired directly from third party supplier (Amphenol p/n = HMC74-0631).

QSFPDD

Agilex 7 FPGA I-Series Transceiver-SoC Development Kit supports 2x QSFPDD ports. QSFDD port fans out from Agilex 7 FPGA I-Series F-Tile (FGT). All 8 channels per QSFPDD can run up to 32G NRZ and 58G PAM4.

Table 14. QSFPDD Connector 0 (13C/J27)

Schematic Signal Name	Description
QSFPDD0_3V3_MODPRS_L	Module present
QSFPDD0_3V3_RESET_L	Module reset
QSFPDD0_3V3_MODSEL_L	Mode select
QSFPDD0_3V3_LPMODE	Initial mode
QSFPDD0_3V3_INT_L	Interrupt
I2C_QSFPDD0_3V3_SCL	I^2C clock
I2C_QSFPDD0_3V3_SDA	I^2C data
QSFPDD0_TX_P/N[0:7]	Transceiver TX
QSFPDD0_RX_P/N[0:7]	Transceiver RX

Table 15. QSFPDD Connector 1 (13A/J48)

Schematic Signal Name	Description
QSFPDD1_3V3_MODPRS_L	Module present
QSFPDD1_3V3_RESET_L	Module reset
QSFPDD1_3V3_MODSEL_L	Mode select
QSFPDD1_3V3_LPMODE	Initial mode
QSFPDD1_3V3_INT_L	Interrupt
I2C_QSFPDD1_3V3_SCL	I^2C clock
I2C_QSFPDD1_3V3_SDA	I^2C data
QSFPDD1_TX_P/N[0:7]	Transceiver TX
QSFPDD1_RX_P/N[0:7]	Transceiver RX

QSFPDD800/MXP

Agilex 7 FPGA I-Series Transceiver-SoC Development Kit supports 1x QSFPDD800 port. QSFDD800 port fans out from Agilex 7 FPGA I-Series F-Tile (FHT). The FHT tile from bank 13A and 13C can run up to 116G PAM4. 4 FHT lanes from bank 13C is terminated directly to QSFPDD800 connector lanes [0:3] (J22).

Table 16. QSFPDD800 (13A)

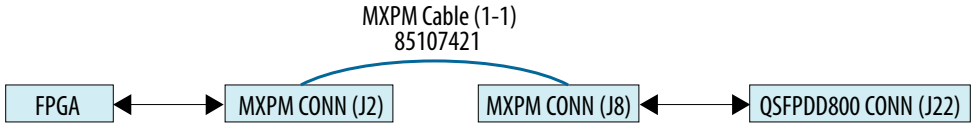
Schematic Signal Name	Description
QSFPDD800_3V3_MODPRS_L	Module present
QSFPDD800_3V3_RESET_L	Module reset
QSFPDD800_3V3_MODSEL_L	Mode select
QSFPDD800_3V3_LPMODE	Initial mode
QSFPDD800_3V3_INT_L	Interrupt
I2C_QSFPDD800_3V3_SCL	I^2C clock

continued...

Schematic Signal Name	Description
I2C_QSFPDD800_3V3_SDA	I^2C data
QSFPDD800_TX_P/N[0:3]	Transceiver TX
QSFPDD800_RX_P/N[0:3]	Transceiver RX

Four FHT lanes from bank 13A are terminated to MXPM connector (J2). Other lanes [4:7] of QSFPDD800 connector is terminated to MXPM connector (J8). For x8 QSFPDD800 topology, 1-1 cable between J2 and J8 need to be used.

Figure 36. MXPM Cable



MXP

MXP port fan out from Agilex 7 FPGA I-Series F-Tile (13A). All four channels can run up to 32G NRZ and 58G PAM4.

Table 17. MXP

Schematic Signal Name	Description
FGT_MXPM_TX_P/N_[0:3]	Transceiver TX
FGT_MXPM_RX_P/N_[0:3]	Transceiver RX

FMC+ Connector

Agilex 7 FPGA I-Series Transceiver-SoC Development Kit supports 2x FMC+ slots for functional expandability. The x16 FGT lanes from bank 12C and 12A are terminated to FMC-A (J7) and FMC-B (J9) connectors, respectively. Auxiliary signals are controlled by the system MAX 10.

SDI Connector

1x FGT channel (13C/Q0/CH3) been terminated to the HD-BNC TX (J1) and RX (J32) connectors to support up to 12G SDI. The system MAX 10 acts as an I^2C master for both SDI driver and receiver, along with control signals.

USB Type-C Connector

Agilex 7 FPGA I-Series Transceiver-SoC Development Kit has hardware support for the USB Type-C connector, which supports DP1.4 specification or USB 3.1 functionality through MUX (TUSB1146). This feature is yet to be validated and implemented.

Serial Buses

SDM I/Os (SDM_IO0/12) and MAX 10 I/Os (VCCL_SDA/SCL) share the same I^2C bus which communicate with Agilex 7 FPGA core regulators. By default, SDM acts as SmartVID master and the system MAX 10 act as Power GUI master in this chain.

The system MAX 10 I/Os (PMB_SDA/SCL) manages the second I2C bus which access all I2C slave regulator except Agilex 7 FPGA core regulators.

The system MAX 10 supports I2C master dedicated to clock-related devices (CLK_I2C_SDA/SCL), which manages 3# clock devices and also connected to the HPS I/Os (HPS_GPIO30/31) through level translator.

Another I2C master instance from the system MAX 10 (VCXO_I2C_SDA/SCL) controls the on board VCXO and Si5394 clock generator.

Agilex 7 FPGA also manages QSFPDD800, 2x QSFPDD, 2DPC DIMM I2C buses, SDI transceivers, and ZL30733 clock synthesizer device.

Table 18. I2C Debug Headers

Schematic Signal Name	Description
PMB_SCL/SDA	VRs I2C header J41
CLK_I2C_SDA/SCL_3V3	System MAX 10 clock I2C bus header J42
ZL_I2C_SDA/SCL	ZL30733 I2C access header J21

Figure 37. I²C Serial Bus (Power Solution 2 Board—non-Enpirion)

This figure applies to DK-SI-AGI027FD, DK-SI-AGI027FC, and DK-SI-AGI027FA.

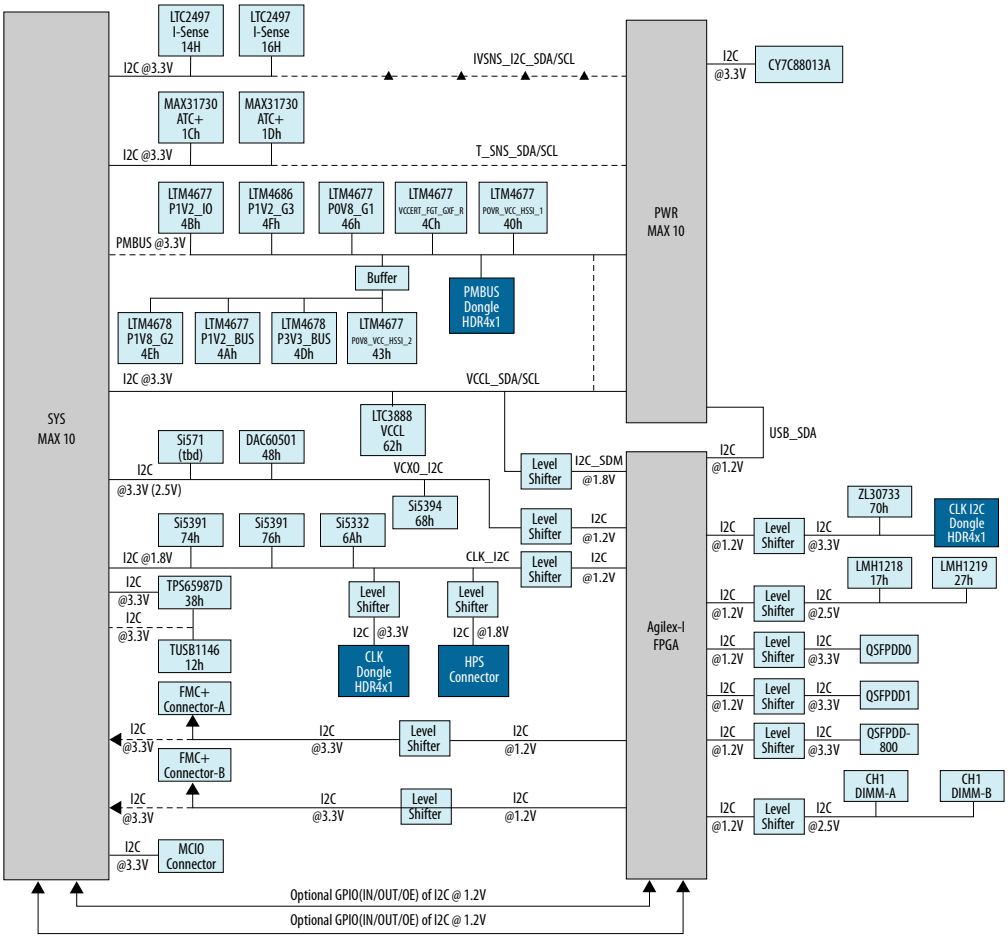
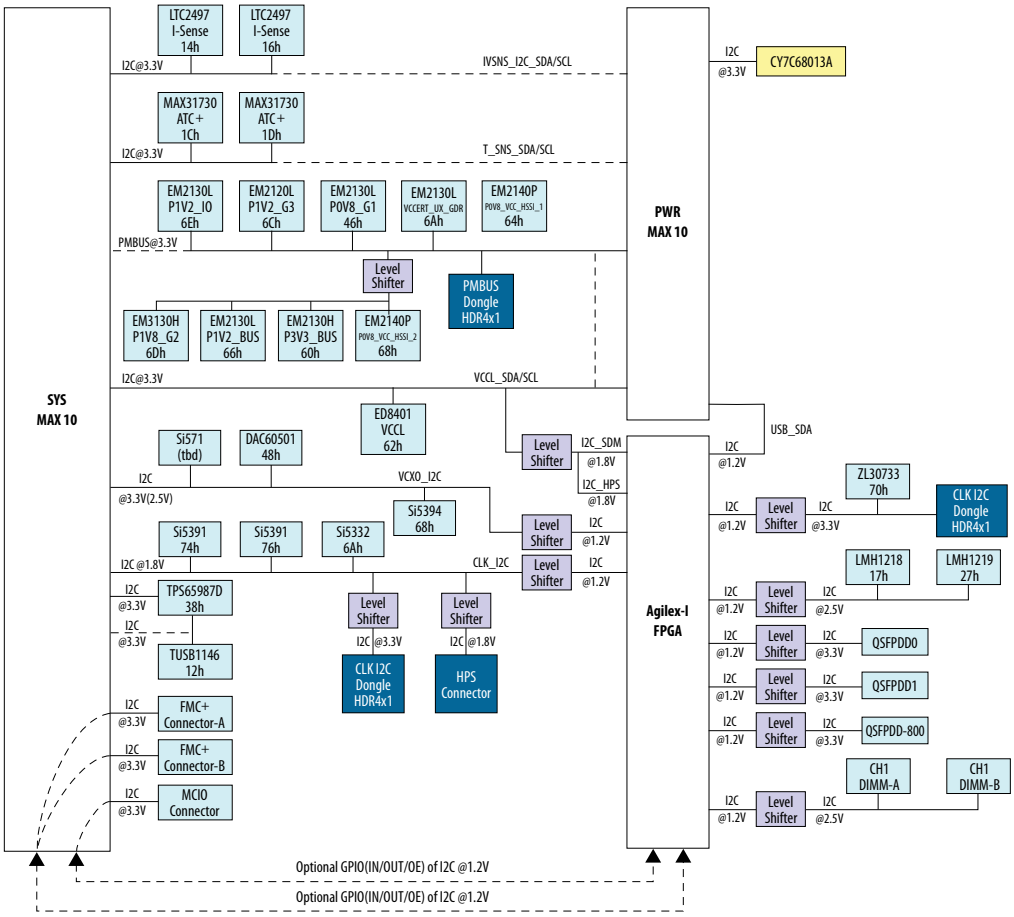


Figure 38. I²C Serial Bus (Power Solution 1 Board—Enpirion)

This figure applies to DK-SI-AGI027FB and DK-SI-AGI027FES.



A.7. Daughter Cards

HPS IO48 OOBE Daughter Card

- 1x RGMII 10/100/1000 Mbps Ethernet port: Standard RJ-45
- 1x UART port: Standard USB Mini-B Receptacle
- 1x Micro SD Card Connector: Standard Micro SD Card Socket
- 1x USB 2.0 port: Standard USB Micro-AB Receptacle
- 1x Mictor 38-pin connector (JTAG only without Trace signals)
- HPS dedicated JTAG pins are connected with both mother board JTAG chain and Mictor 38-pin header

- I^2C: HPS I^2C port
- GPIO
  - ☐ 2x Push buttons
  - ☐ 3x LEDs
  - ☐ 1x Ethernet Interrupt from Ethernet PHY
  - ☐ 1x USB over-current indicator
- HPS Clock: 25 MHz oscillator

256 MB QSPI Flash Daughter Card

This daughter card is pre-programmed with GHRD for AS configuration. It can be re-programmed by user image. The part number is MT25QU02GCBB8E12-0SIT.

A.8. Connectors and Cables

Table 19. Connectors and Cables

Connectors and Cables	Part Number	Switch Reference	Notes
QSFPDD loopback module	NLNAMB0001 (Amphenol)	J27 J48 J22 (for use up to 400 G)	—
QSFP-DD800 loopback type 1	ML4062-LB-112 (Multilane)	J22	For 800 G usage, connect 85107421 at J2 and J8
MXMP breakout cable	85107421 (Huber + Suhner)	J2–J8	Connect between J2 and J8
MCIO cable	HMC74-0631 (Amphenol)	J37	—
HD-BNC Straight Plug to HD-BNC Straight Plug	095-850-214M100 (Amphenol)	J1 J32	SDI loopback between J1 and J32

B. Developer Resources

Use the following links to check the Altera website for other related information.

Table 20. Agilex 7 FPGA I-Series Transceiver-SoC Development Kit References

Reference	Description
Agilex 7 FPGA I-Series Transceiver-SoC Development Kit (4x F-Tile) webpage	Latest board design files, reference designs, and kit installation for Windows* and Linux*.
Altera FPGA Developer Site	Open-source community website supporting SoC development including Altera and Partner SoC development kit targets and related designs and documentation.
Agilex 7 Design Hub	The Agilex 7 Design Hub is a comprehensive resource hub that provides a structured approach for developing FPGA-based platforms.
AN 958: Board Design Guidelines	Board design-related resources for Altera devices. Its goal is to help you implement successful high-speed PCBs that integrate device(s) and other elements.
Agilex 7 Power Management User Guide	Describes the Agilex 7 devices power-optimization features, power-up and power-down sequences, power distribution network, voltage and temperature monitoring systems with a design example to read the TSDs, and power optimization techniques.
Agilex 7 Power Distribution Network Design Guidelines	Provides information for the Agilex 7 device family power distribution network (PDN) design guidelines.
FPGA SmartVID	SmartVID is a feature on select Altera FPGAs where the device identifies the optimal voltage that it should be operated at, and provides this information to the power regulator via the PMBus. The term represents Smart Voltage IDentification (SmartVID).
SmartVID Debug Checklist and Voltage Regulator Guidelines	Provides the checklist to assist you to rule out the possible causes of configuration failure due to SmartVID.
Agilex 7 Configuration User Guide	Provides the configuration process, the device pins required for configuration, the available configuration schemes, remote system updates, and debugging. This user guide also provides an overview of the secure device manager (SDM) which manages security for the configuration bitstream.
Documentation: Agilex 7	Agilex 7 device documentation.
Cadence* Capture CIS Schematic Symbols	Agilex 7 OrCAD symbols.

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*Other names and brands may be claimed as the property of others.

C. Safety and Regulatory Compliance Information

C.1. Safety and Regulatory Information



ENGINEERING DEVELOPMENT PRODUCT - NOT FOR RESALE OR LEASE

This development kit is intended for laboratory development and engineering use only.

This development kit is designed to allow:

- Product developers and system engineers to evaluate electronic components, circuits, or software associated with the development kit to determine whether to incorporate such items in a finished product.
- Software developers to write software applications for use with the end product.

This kit is not a finished product and when assembled may not be resold or otherwise marketed unless all required Federal Communications Commission (FCC) equipment authorizations are first obtained.

Operation is subject to the condition that this product not cause harmful interference to licensed radio stations and that this product accept harmful interference.

Unless the assembled kit is designed to operate under Part 15, Part 18 or Part 95 of the United States Code of Federal Regulations (CFR) Title 47, the operator of the kit must operate under the authority of an FCC licenseholder or must secure an experimental authorization under Part 5 of the United States CFR Title 47.

Safety Assessment and CE & UKCA mark requirements have been completed, however, other certifications that may be required for installation and operation in your region have not been obtained.

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*Other names and brands may be claimed as the property of others.

C.1.1. Safety Warnings



Power Supply Hazardous Voltage

AC mains voltages are present within the power supply assembly. No user serviceable parts are present inside the power supply.

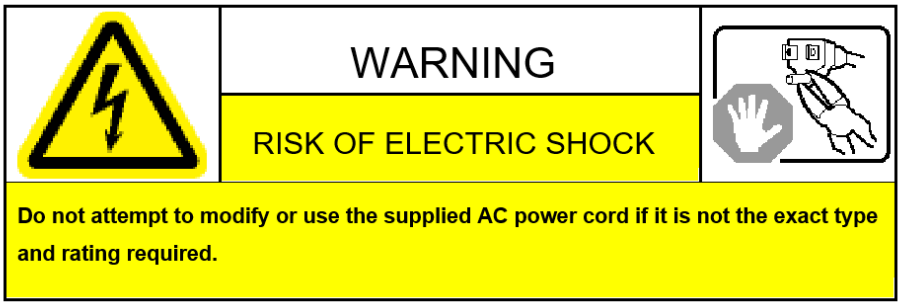
Power Connect and Disconnect

The AC power supply cord is the primary disconnect device from mains (AC power) and used to remove all DC power from the board/system. The socket outlet must be installed near the equipment and must be readily accessible.



System Grounding (Earthing)

To avoid shock, you must ensure that the power cord is connected to a properly wired and grounded receptacle. Ensure that any equipment to which this product is attached to is also connected to properly wired and grounded receptacles.



Power Cord Requirements

The plug on the power cord must be a grounding-type male plug designed for use in your region. It must have certification marks showing certification by an agency in your region. The connector that plugs into the appliance inlet of the power supply must be an IEC 320, sheet C13, female connector. If the power cord supplied with the system does not meet requirements for use in your region, discard the cord, and do not use it with adapters. Use only certified power supply cord with appropriate gauge, designed for use in your region.



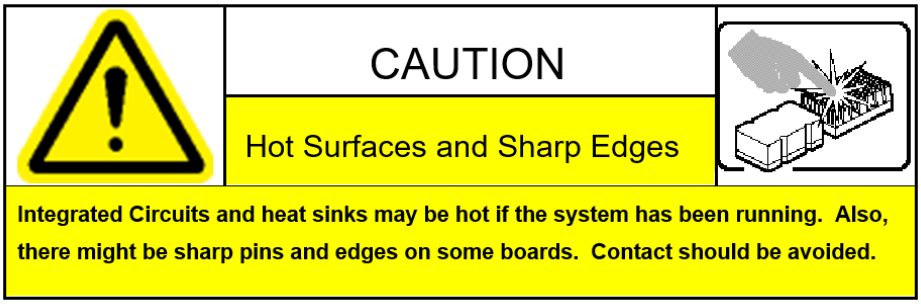
Lightning/Electrical Storm

Do not connect/disconnect any cables or perform installation/maintenance of this product during an electrical storm.

Risk of Fire

To reduce the risk of fire, keep all flammable materials a safe distance away from the boards and power supply. You must configure the development kit on a flame retardant surface.

C.1.2. Safety Cautions



Thermal and Mechanical Injury

Certain components such as heat sinks, power regulators, and processors may be hot. Heatsink fans are not guarded. Power supply fan may be accessible through guard. Care should be taken to avoid contact with these components.





Cooling Requirements

Maintain a minimum clearance area of 5 centimeters (2 inches) around the side, front and back of the board for cooling purposes. Do not block power supply ventilation holes and fan.

Electro-Magnetic Interference (EMI)

This equipment has not been tested for compliance with emission limits of FCC and similar international regulations. Use of this equipment in a residential location is prohibited. This equipment generates, uses and can radiate radio frequency energy which may result in harmful interference to radio communications. If this equipment does cause harmful interference to radio or television reception, which can be determined by turning the equipment on and off, you are required to take measures to eliminate this interference.

Telecommunications Port Restrictions

The wireline telecommunications ports (modem, xDSL, T1/E1) on this product must not be connected to the Public Switched Telecommunication Network (PSTN) as it might result in disruption of the network. No formal telecommunication certification to FCC, R&TTE Directive, or other national requirements have been obtained.



Electrostatic Discharge (ESD) Warning

A properly grounded ESD wrist strap must be worn during operation/installation of the boards, connection of cables, or during installation or removal of daughter cards. Failure to use wrist straps can damage components within the system.

Attention: Please return this product to Altera for proper disposition. If it is not returned, refer to local environmental regulations for proper recycling. Do not dispose of this product in unsorted municipal waste.

Lithium Ion Battery Warnings



Lithium Battery: Risk of explosion if the lithium battery is replaced by an incorrect type. Risk of fire, explosion, or chemical burn if the battery is mistreated (punctured or crushed). Do not attempt to disassemble. Do not incinerate. Observe proper polarity when replacing battery. Do not dispose—the battery is intended to be serviced and disposed by qualified Altera service personnel only.

Perchlorate Material: Special handling may apply. For more details, refer to www.dtsc.ca.gov/hazardouswaste/perchlorate. This notice is required by California Code of Regulations, Title 22, Division 4.5, Chapter 33: Best Management Practices for Perchlorate Materials. This product includes a battery which contains perchlorate material.

Taiwan battery recycling:



(Translation - please recycle batteries)

Please return this product to Altera for proper disposition. If it is not returned, refer to local environmental regulations for proper recycling. Do not dispose of product in unsorted municipal waste.

C.2. Compliance Information

CE EMI Conformity Caution

This development board is delivered conforming to relevant standards mandated by Directive 2014/30/EU. Because of the nature of programmable logic devices, it is possible for the user to modify the development kit in such a way as to generate electromagnetic interference (EMI) that exceeds the limits established for this equipment. Any EMI caused as a result of modifications to the delivered material is the responsibility of the user of this development kit.

